

Received 23 June 2025, accepted 2 July 2025, date of publication 10 July 2025, date of current version 16 July 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3587231

SURVEY

A Systematic Literature Review of Concept Drift Mitigation in Time-Series Applications

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This work was supported in part by the Universiti Teknologi PETRONAS under Grant YUTP-FRG (015LC0-487).

ABSTRACT Machine Learning (ML) plays a key role in time-series applications because it analyzes observed data and predicts future values. The effectiveness of ML models in time-series forecasting is reduced by the occurrence of Concept Drift (CD). CD refers to continuous changes in the statistical properties of datasets. This affects the predictive performance of ML models. To address this issue, researchers have developed several CD detection and adaptation techniques. However, certain limitations exist, and most studies on CD detection and adaptation in time-series data have focused on classification learning, with minimal attention paid to regression learning. This study conducted a Systematic Literature Review (SLR) that classified, identified, and recommended an optimal method for the detection and adaptation of CD in regression and classification tasks involving time-series data. A systematic search was performed using the SCOPUS, ScienceDirect, IEEE Xplore, Web of Science, MDPI, and ACM databases. Based on the identified records, 60 studies published between 2013 and 2024 were thoroughly surveyed and evaluated using PRISMA guidelines. The findings show that Support Vector Machines (SVM) is the most effective learning algorithms for the detection and adaptation of CD in regression and classification tasks using time-series data. This is possible because of their high detection accuracy and effective memory. Moreover, this SLR presents a roadmap for detecting CDs using Artificial Intelligence (AI)-based learners, along with a comparative analysis of well-known baseline methods. Future work should focus on developing adaptive learning models capable of adapting to CD without explicit re-training, thereby, ensuring optimal performance.

INDEX TERMS Systematic literature review, concept drift, time series, data stream, regression, nonstationary environment, machine learning.

I. INTRODUCTION

In real-world applications, time-series data represent any set of recorded observations. Significant applications of time-series data in various fields, such as energy

The associate editor coordinating the review of this manuscript and approving it for publication was Chao Tong.

consumption, have increased in popularity in recent decades [1]. Additionally, time-series data are used for analyses in various domains such as financial variables, astronomy, energy, and weather forecasting [2]. Recent technological advancements have led to significant progress in data collection, facilitating the accumulation of large amounts of time-series data over time [3]. The phenomenon

known as Concept Drift (CD) signifies a change in the relationship between input data and model target values. CD continuously changes the statistical properties of a dataset over time, which affects the performance of Machine Learning (ML) models, particularly after deployment. For example, a comparative analysis of an artificial time-series with a simulated CD problem demonstrated the presence of changing data over time. Three artificial time-series exhibiting CD were generated by combining data from two autoregressive (AR) processes. Each generated time-series had 20,000 instances, with CD occurring at 10,001 [4]. In addition, CD occurs when a target concept encounters changes in a nonstationary environment [5]. The CD problem can reduce the efficacy of ML models trained using these data. Therefore, consistent monitoring of ML applications throughout the development lifecycle is essential to address systematic fluctuations in the data distribution.

Detecting CD in time-series data ensures the reliability of the predictive model in dynamic contexts. Various techniques such as ML, statistical methods, and adaptive combination ensemble learning have been proposed to address the CD problem [6], [7], [8]. For instance, in the health field, a ML model has been used to evaluate the impact of data drift on clinical sepsis prediction, emphasizing the necessity of continuous model monitoring and updating to maintain predictive accuracy [9]. In addition, a transfer learning method has been used to address temporal performance drift over time in clinical prediction models [10]. A similar study proposed a hybrid early drift detection method (HEDDM) that uses statistical distance measurements to efficiently detect CD. In addition, it improves the accuracy of drift detection in data streams [11]. However, several adaptive combinations and ensemble-based learning algorithms have demonstrated the ability to detect CD in multiple domains. An adaptive ensemble learning method was presented to address CD detection, thereby improving the performance of intrusion detection systems in dynamic network environments [12].

Despite these advancements, previous studies on CD detection and adaptation have several limitations. Studies have primarily focused on classification learning, with minimal attention paid to regression learning tasks. Several issues remain in the efficient identification of changes in data distribution features and responses to CD in a time-series context in the contemporary literature. Additionally, the evaluation metric performance for the validation of CD in ML models using benchmark datasets may not accurately reflect real-world industrial data. Determining an appropriate CD detection technique for various datasets is challenging. Addressing these limitations can enhance ML performance and computational efficiency.

This study systematically examined state-of-the-art CD detection techniques in time-series applications, highlighting their strengths and their limitations. The SLR began with an in-depth investigation of the issue of CD in a time-series context for both ML classification and regression learning algorithms. A comparative review of existing CD detection

approaches based on accuracy, adaptability, and computational efficiency was conducted by synthesizing recent developments and identifying research gaps. Additionally, the SLR posed and answered four research questions based on existing empirical studies on drift detection and adaptation. To the best of our knowledge, this is one of the first comprehensive SLR to investigate the detection and adaptation of CD in regression and classification tasks involving time-series applications.

The main contributions of this study are twofold: First, it provides a comprehensive review of the most effective learning algorithms for detection and adaptation to CD in regression and classification tasks using time-series data. Many surveys and reviewers have addressed the issue of CD in ML classification learning tasks; however, this remains a challenge. This study investigated the CD problem for both classification and regression learning tasks using time-series data. Second, this study provides a roadmap for detecting CD using AI-based learners, the limitations of the study, and potential future recommendations.

A. CONCEPT DRIFT OVERVIEW

This section provides a background on CD mitigation, including methods for time-series applications. As CD continuously changes the statistical properties of a dataset over time, it deteriorates ML performance. In addition, changes in the statistical properties of the data CD or the relationships between target variables during training and testing (data shift) affect the ML models. CD can be classified according to their speed as incremental, gradual, sudden, or recurring (Figure 1). An incremental CD occurs when there is a minimal change in the original data distribution, and a gradual CD occurs when there are noticeable changes in the target data distribution. Sudden CD refers to a significant change in the original data distribution at a specific time point, whereas recurrent CD refers to a situation in which an old concept reappears after a period of absence. An identical concept called dataset shift in the context of time-series data represents any change in the joint distribution of input and target variables between the training and testing stages. Dataset shifts encompass several types including concepts, covariates, and prior probability shifts. The key difference between CD and dataset shift is that CD is inherently dynamic and refers to real-time changes in the data distribution. In contrast, a dataset shift frequently refers to discrepancies that exist between different datasets, such as training or testing, but does not always imply time evolution [13]. The combined CD and dataset shift problem significantly affects the ML development life cycle.

In addition, time-series intelligent application integrity relies on the ML development life cycle, where the phenomenon ‘CD’ interrupts the process by changing the data distribution, which leads to a decrease in model performance. This has attracted the attention of many researchers who have developed methods to address CD problems in time-series data [15], [16], [17]. Several studies have investigated

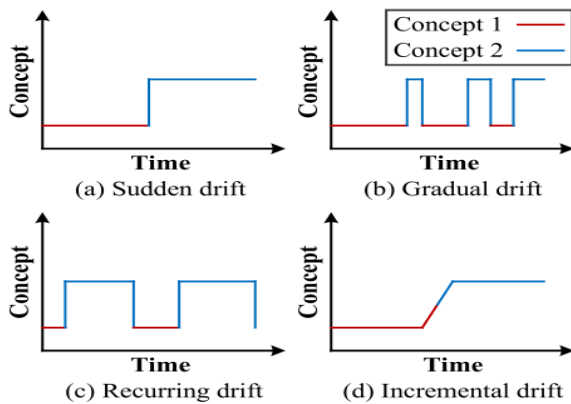


FIGURE 1. Concept drift types based on speed [14].

CD using time-series data and workable solutions. Uchitel-eva et al. [18], for example, introduced a new approach that is sensitive to drift for modeling multivariate time-series data in nonstationary environments of Industrial Internet of Things (IIoT). The three-layered, three-state (TriLS) system enables collaboration between the gateway and cloud to adjust the lightweight predictive model in time.

This SLR consists of eight sections and is organized as follows: Section I provides an introduction and background on CD mitigation, including its methods in time-series applications. Section II presents a comprehensive review of the literature on CD detection methods. Section III presents the survey methodology based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. Section IV presents the results, including an in-depth summary of screening and search procedures. Section V presents the major SLR findings based on the research questions. Section VI presents a taxonomy that provides a brief overview of advances in the handling, experimental evaluation, and comparative analysis of CD. Section VII discusses the study findings. Section VIII presents summary of lessons learned and best practices. Section IX. presents conclusions, limitations, and directions for future research.

II. RELATED REVIEWS

This section presents a comprehensive review of the literature on CD detection. Drift problems occur when a relationship changes over time, affecting prediction accuracy. Different CD methods exist, such as data distribution, performance-based methods, and multiple hypotheses. For example, data-distribution-based methods monitor changes in fundamental data distribution, which may indicate a shift in the underlying patterns and relationships. These methods can be applied to early financial forecasting and network intrusion detection. Performance-based techniques detect CD by evaluating model accuracy or error rates over time. Multiple hypothesis methods are commonly employed in CD detection for

predictive maintenance tasks, such as autonomous vehicles and recommendation systems, in which adapting to evolving data distributions is critical. These approaches maintain a set of competing models, allowing dynamic selection of the most relevant hypothesis based on the current input. This strategy is particularly important in areas such as fraud detection and healthcare diagnostics, where both gradual and sudden changes require strong and adaptable detection methods to ensure predictive accuracy. This study aimed to identify the most efficient method for detecting CD in a time-series setting. Due to the CD issue, it has been increasingly difficult to predict time-series using ML outputs after training the data. However, the effectiveness of models trained on historical data can decrease over time owing to these changes.

Bayram et al. [14] conducted a comprehensive analysis and categorization of performance-based CD detection methods proposed over the past decade. These methods leverage the decline in performance of the predictive model as an indicator of significant system alterations. The authors also used a hierarchical diagram to present a systematic organization of the classification, facilitating a structured and coherent navigation among various methods. A similar study by Shen et al. [19] presented a comprehensive survey of current unsupervised CD detection methods. Additionally, this study provides a thorough overview and innovative taxonomy of unsupervised CD detection approaches for real applications, highlighting their effectiveness in identifying appropriate model update timing amid evolving data distributions; however, it lacks an empirical evaluation. Han et al. [20] presented a survey that introduced the concept of CD to nonstationary data streams. CD poses significant challenges in various domains such as data mining and ML. This study introduces a novel approach to classifying these methods and provides a comprehensive explanation of the current methods for managing CD, including both active and passive methods. Gemaque et al. [21] investigated various techniques to handle CD in unsupervised classification tasks. An additional contribution is the proposal of a taxonomy that categorizes state-of-the-art CD detection approaches focusing specifically on unsupervised strategies. The authors also conducted a survey and analysis of various approaches to CD. Nayak et al. [22] presented an overview of the research field of CD and a comprehensive review of the diverse approaches employed by researchers to address the challenges associated with CD. In addition, it emphasizes ensemble learning, adaptive windowing, and weighting techniques as essential methods for sustaining model performance, despite changing data distributions. However, this study did not provide insights into unsupervised or semi-supervised contexts. Another study by Poenaru-Olaru et al. [23] evaluated the efficacy of CD detectors in detecting temporal drift by examining the timeliness of drift reports and frequency of generated false alarms. The authors also conducted a comparative analysis of the performance of prominent drift detectors, which were categorized into two distinct groups: error-rate-based detectors and data-distribution-based detectors.

TABLE 1. Comparison with other studies in the same domain (✓ : Yes).

Authors	Year	Time-series Data	Systematic Review	ML Classification	ML Regression
Bayram et al. [14]	2022			✓	
Shen et al. [19]	2023			✓	
Han et al. [20]	2022	✓		✓	
Gemaque et al. [21]	2020			✓	
Nayak et al. [22]	2021			✓	
Poenaru-Olaru et al. [23]	2022			✓	
Aminikhanghahi et al. [29]	2017	✓		✓	
Wang et al. [30]	2018		✓	✓	
Barros et al. [31]	2019			✓	
Ghani et al. [32]	2020		✓	✓	
Gemaque et al. [21]	2020			✓	
Nayak et al. [22]	2021			✓	
Mahmood et al. [33]	2021		✓	✓	
Lima et al. [34]	2022		✓		✓
Desale et al. [35]	2022			✓	
Ghatage and Patil [36]	2023	✓			
Suárez-Cetrulo et al. [37]	2023			✓	
Goncalves et al. [38]	2024		✓	✓	
Mohandas et al. [39]	2024			✓	
This study	2025	✓	✓	✓	✓

In addition, this study considers studies that attempt to mitigate CD problems using a substantial body of literature on domain adaptation. For example, Karimian and Beigy [24] proposed a method for addressing CD in data streams using a domain-adaptation technique. The authors proposed a technique called Concept Drift Domain Adaptation (CDDA), which effectively addresses CD using multi-source domain adaptation. The model efficiency was enhanced in terms of execution time and memory usage by adopting this technique. These studies also include a thorough theoretical analysis of CDDA and present a generalization bound that demonstrates the problem of predicting the data streams. The proposed method was tested using both synthetic and real-world data streams. A similar study by Gama et al. [25] provided a comprehensive overview of adaptive learning methods for handling CD problems in dynamic data environments. This study also categorized and evaluated various strategies, including window-based approaches and ensemble methods, along with their strengths and limitations. The survey is well-structured, covering theoretical foundations, evaluation methodologies, and practical applications in various domains.

Taxonomic details and comparisons of existing techniques highlight the complexity and interdisciplinary nature of the problems and provide useful guidance for researchers and practitioners.

Due to certain limitations in the existing literature, most studies have focused on classification learning with minimal attention to regression learning within the context of time-series. This review focuses on the applications of CD detection techniques between 2013 and 2024. However, before 2013, drift detection was accomplished using several approaches and conventional procedures. The methodologies employed during this period included statistical hypothesis testing, classification error distance measurement, Enhanced Customer Due Diligence (ECDD), and online MapReduce. Hoens et al. [26] provided a comprehensive overview of various CD domains in 2012. Each of these areas is discussed in detail, highlighting the difficulties involved. In 2009, Guzella and Caminhas [27] presented a broad overview of CD detection using ML algorithms for spam filtering. In 2007, Gaber et al. [28] presented a comprehensive survey on CD detection and the challenges associated with the

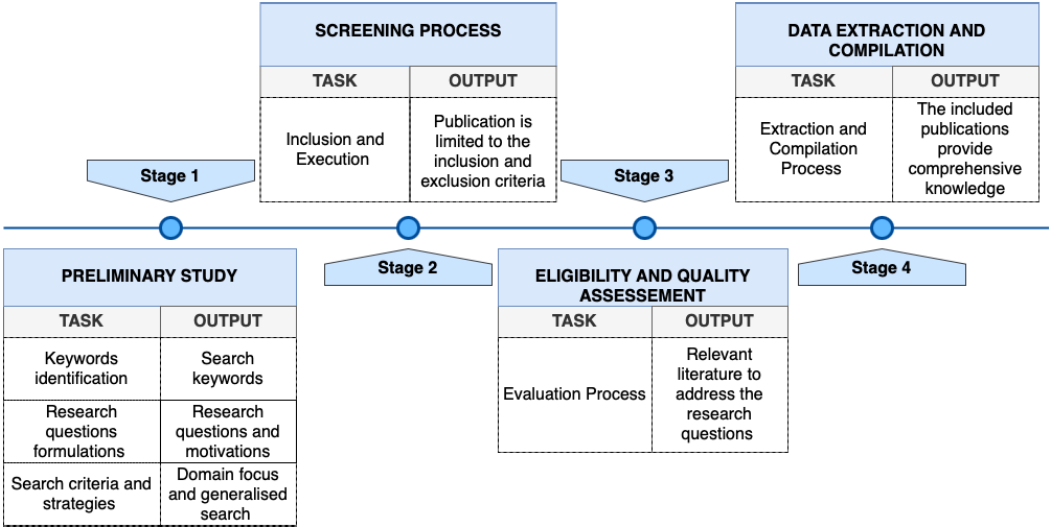


FIGURE 2. Systematic literature review mapping process.

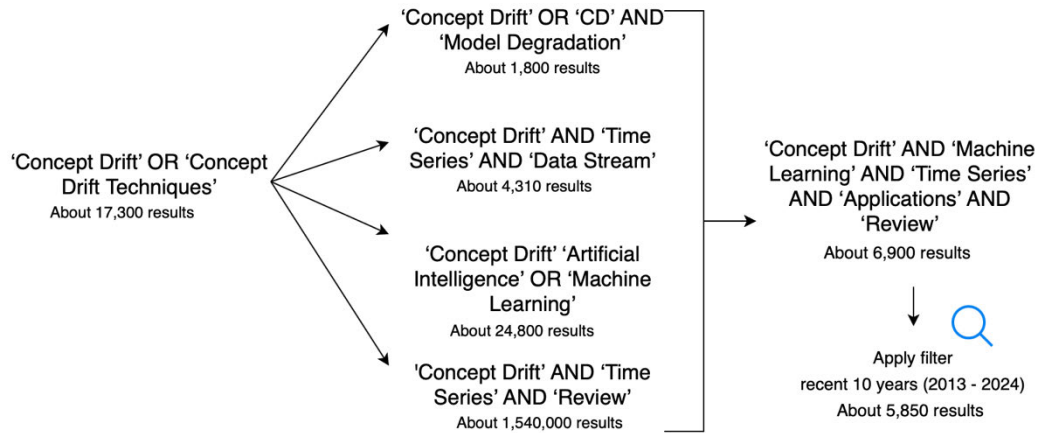


FIGURE 3. Google Scholar keyword combination results.

data-stream domain. Furthermore, the methodologies used for the problem necessitate a comprehensive redesign to effectively tackle the challenges posed by the limited resources and CD phenomenon. A summary of the additional research related to these domains is presented in Table 1.

III. SURVEY METHODOLOGY

This systematic literature review was performed using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement [40], [41]. The study screening and selection processes followed the standard guidelines established by Kitchenham and Charters [42]. This review aims to present classification and regression learning approaches while investigating the most efficient methods for detecting and adapting CD in time-series and data streams. To ensure comprehensive reporting, the structure and

methodology of previous in-depth studies were adopted [43], [44], [45]. Figure 2 shows the SLR mapping process for the survey methodology section of this study, which encompasses four stages in the screening process.

A. STAGE 1: PRELIMINARY STUDY

This section presents the preliminary study. Consequently, an initial investigation was conducted to obtain a thorough understanding. This step motivated the authors to develop robust RQs, keywords, and study parameters for the review. Three tasks were performed during the preliminary stage: keyword identification (1), research question (RQ) formulation (2), and search criteria and strategy (3).

1) KEYWORD IDENTIFICATION

The search strategy for keyword identification was developed for six databases using the following search terms: 'Concept

TABLE 2. Formulated research questions and motivation.

ID	Question	Motivation
RQ1	Which ensemble-based learning methods are most effective for detecting CD in time-series data using classification?	To investigate the most effective ensemble-based learning method for classification and regression to detect CD in a time-series environment in which data characteristics are constantly changing.
RQ2	What are the various ML models used in a general context to detect CD, and how do they perform in terms of actionability in drift detection and adaptation tasks?	To investigate existing ML models in a general context and provide a comprehensive roadmap for future research and development by combining existing research and identifying knowledge gaps.
RQ3	What algorithms have been used in the last decades for analyzing and handling CD in time-series data and data streams?	To explore the existing algorithms learners used for analyzing and handling CD in a time-series environment.
RQ4	What are the main steps to address CD problems, specifically in time-series data and the general context?	To identify a general pipeline for detecting CD in both time-series data and general contexts.

Drift’ OR ‘CD’ AND ‘Model Degradation’ AND ‘Drift Handling’ OR ‘Drift Detector’ AND ‘Concept Evaluation’ AND ‘Time-Series’ OR ‘Time-Series Data’ AND ‘Artificial Intelligence’ OR ‘AI’ AND ‘Machine Learning’ OR ‘ML’ AND ‘Regression’ AND ‘Classification.’ These keywords were selected to represent CD research. This study conducted a preliminary search of the Google Scholar database using the keyword ‘Concept Drift’ to explore the existing literature on this topic. Figure 3 shows the results based on different keyword combinations and the number of papers obtained through Google Scholar search. The initial search output was reduced to 5,850 after the quality of the retrieval results was evaluated using filtering. In addition, various synonyms for “Concept Drift” were obtained from Lima et al. [34]. All journal articles published in English were included in all searches covering 2013 – 2024 in the selected databases.

2) RESEARCH QUESTIONS FORMULATION

Authors (M.A. and Y. B.) collaborated to describe the RQs and their motivations for this study based on a literature review. Table 2 presents the research questions and their motivations.

3) SEARCH CRITERIA AND STRATEGIES

Search criteria and strategies are crucial when searching for specific domains. Six databases were used to search for selected literature: SCOPUS, Science Direct, IEEE Xplore, Web of Science, ACM Digital Library, and MDPI. A search was conducted using titles, abstracts, and keywords to identify studies relevant to CD. The selected publication types included journals and conference proceedings published between 2013 and 2024 covering decades. To refine the search, a comprehensive search query was developed and used to examine the titles, abstracts, and keywords in the literature. The domain focus is CD the keywords are ‘Concept drift, time-series’ the search method is ‘Title, Abstract, Keywords’ and the search strings are ‘Concept Drift’ OR ‘CD’ AND ‘Model Degradation’ AND ‘Drift Handling’

TABLE 3. Inclusion and exclusion criteria.

Criteria	Inclusion	Exclusion
The type of study	Original review articles.	Book chapters, thesis, reports, newsletters, and editorials.
Source	Peer review and conference proceedings.	
Publication year	January 2013-2024.	Pre 2012-Post February 2024.
Language	English.	Non-English.
Region	Not restricted to a particular region.	
Relevance	Relevant to the developed RQs	Not relevant to the developed RQs
Settings	CD in the time-series context.	Non-time-series context.

AND ‘Time-series.’ Figure 4 depicts the screening and selection process using PRISMA 2020 standards, where relevant publications were identified.

B. STAGE 2: SCREENING PROCESS

The database search, which used identified keywords and a generalized search string, yielded 183 unique papers that were reviewed thoroughly. The screening process of the articles was evaluated by two authors (S.J.A. and N.A.).

1) INCLUSION AND EXCLUSION CRITERIA

This study systematically reviewed the available literature on CD detection and time-series data in the fields of mathematics, computer science, engineering, and decision science. The search also encompassed the period from 2013 to 2024, except for articles published earlier in 2012. In addition, rather than being limited to a single nation or area, a search was conducted worldwide. After excluding 123 articles, 60 research articles were selected. The criteria used to select the research articles for inclusion and exclusion are presented in Table 3.

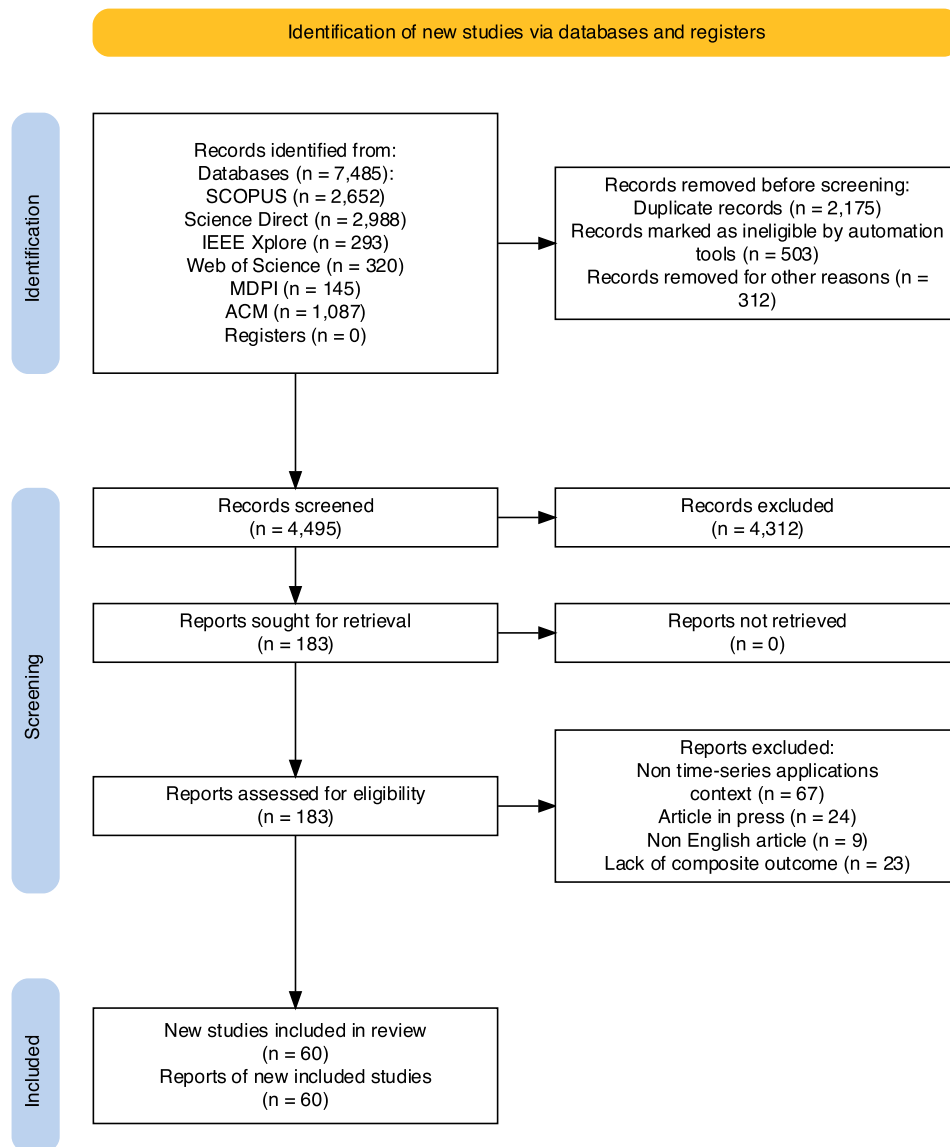


FIGURE 4. PRISMA 2020 flow diagram of study screening and selection.

C. STAGE 3: ELIGIBILITY AND QUALITY ASSESSMENT

Eligibility and quality assessments were conducted using reviews and primary research. Each duplicate record was carefully examined to ensure review quality. In particular, the abstracts of research articles that were part of the review process were thoroughly examined and screened to determine their quality and significance. The subsequent step involved in-depth evaluation of each research article. Another exclusion criterion was restricting the articles to those published in English. Hence, eight articles written in languages other than English were excluded.

After screening, 60 papers were evaluated for eligibility and quality. The evaluation was performed using specific criteria and corresponding scoring values (Table 4). A score

of 1 was assigned for “Yes,” 0.5 for “Partially” and 0 for “No.” The eligibility and quality assessment scoring criteria were validated based on the selected studies. The decision was made after excluding studies that failed to meet the scoring criteria; however, all studies that received a score of 1 or 0.5 were included in the data extraction process related to the RQs. However, this was not performed carelessly because the study followed a thorough evaluation of eligibility and quality. The effectiveness of the assessment proved valuable in examining existing related studies, specifically those focusing on their objectives and methodologies for mitigating CD in time-series environments. This evaluation ensured that the final selection consisted only of studies with clearly defined objectives, thoroughly explained methodologies, acknowledged limitations, and clear conclusions.

TABLE 4. Eligibility and quality assessment scoring criteria.

Criteria	Score	Description
Are the research objectives well-defined?	1	Yes, the study explicitly defines its objectives.
	0.5	The study defines its objectives; however, its goals are inadequately defined.
	0	No, the study lacks clearly defined objectives.
Are the research methodologies well-defined?	1	Yes, the study includes a comprehensive method.
	0.5	The proposed method requires further detailed investigation.
	0	No, the study lacks a well-defined method.
Does the study describe the challenges associated with CD detection and adaptation?	1	Yes, the study examines the challenges related to the detection and adaptation to CD.
	0.5	The study discusses the challenges associated with detecting only CD.
	0	No, the study did not discuss the difficulties associated with CD detection and adaptation.
Does the study provide a clear description of the dataset used to detect CD?	1	Yes, the study provides a clear description of the datasets used to detect CD and adaptation.
	0.5	The study used only artificial datasets to detect CD.
	0	No, the study did not provide the dataset details used to detect CD.
Does the study accurately represent the research findings?	1	Yes, the study presents clear findings and suitable data visualization.
	0.5	The study findings lack clear statements or visualizations.
	0	The findings of the study are unclear and lack detailed explanations.

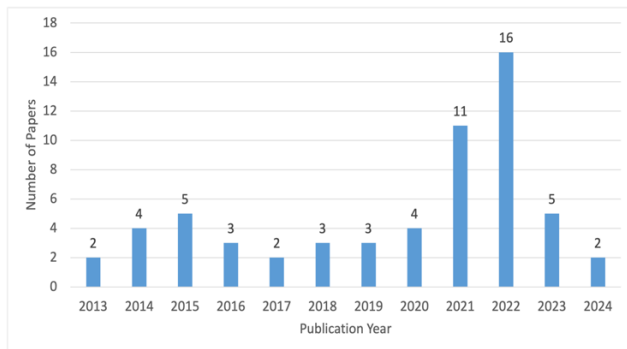


FIGURE 5. Distribution of studies by the publication year.

D. STAGE 4: DATA EXTRACTION AND COMPILATION

After completing the eligibility and quality assessments, the selected studies were entered for data extraction using Microsoft Excel (version 16.77). The data extracted from the selected studies included the publication year, algorithm used, metrics evaluation, domain, and ML techniques. However, data extraction was conducted by two authors (S.J.A. and N.A.).

In addition, the extracted data included authors, title, abstract, volume, number, pages, publisher, keywords, year, datasets, CD algorithm, evaluation metrics, problem scope, and algorithm learning. Finally, 60 studies were included and systematically analyzed to address the RQs of this review. These studies provide useful insights into the detection of CD in time-series applications.

IV. RESULTS AND ANALYSIS

This section presents an in-depth summary of the findings of the screening and search procedures based on the PRISMA guidelines. First, the study provides a quantitative description

of the data from a selected group of research studies, describing the characteristics according to publication year, journal, problem scope of ML, evaluation metrics, and learning algorithms. Furthermore, this study categorized the selected studies using learner algorithms, evaluated model performance, and employed techniques that effectively addressed CD.

A. CHARACTERISTICS OF THE SELECTED STUDIES

Figure 5 presents the number of publications on CD using time-series applications from 2013 to 2024 and highlights the growing body of research on this topic. The findings indicate that the number of published papers has increased over time, with a particularly significant increase over the last few decades. Additionally, the distribution of studies shows a growing emphasis on CD research, as evidenced by the significant increase in publications from 2021 to 2022, with the highest number of publications occurring in 2022. This high number can be attributed to the concentration of CD issues in response to practical challenges encountered in real-world applications, especially in sectors where model reliability over time is crucial, including finance, healthcare, and energy. The results demonstrate the importance of ensuring the robustness and relevance of predictive ML models in time-series environments. The highest temporal number increases in 2021 and 2022 may represent advancements in CD detection and adaptation methodologies, which can be attributed to advances in ML model development. Finally, the growing interest in CD can be attributed to the increasing complexity of time-series data and data streams, as well as increased awareness of the potential problems that CD can cause.

Figure 6 shows the distribution of studies classified according to specific journals, where (N = 46) were published in journals and the remaining (N = 14) were published in

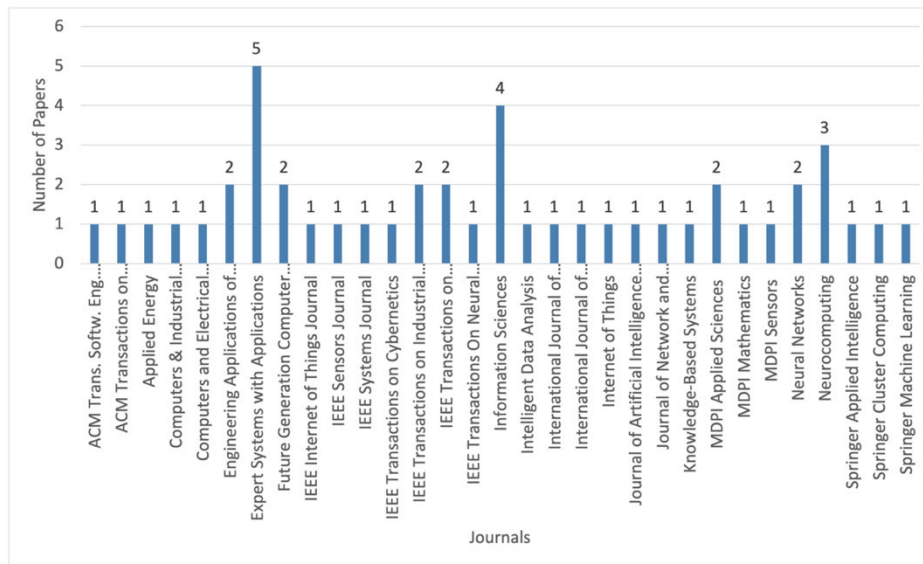


FIGURE 6. Distribution of studies by journals.

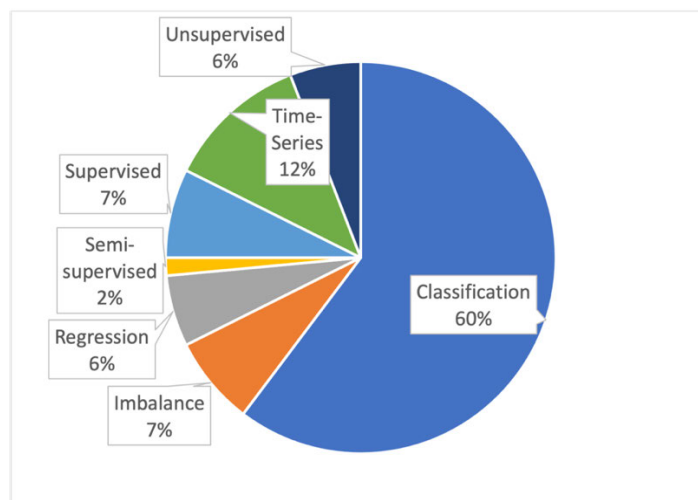


FIGURE 7. Characteristics of studies based on the problem scope of machine learning.

conference proceedings. Journal Source: *ACM Trans. Softw. Eng. Methodol.*, *ACM Transactions on Intelligent Systems and Technology*, *Applied Energy*, *Computers & Industrial Engineering*, *Computers and Electrical Engineering*, *Engineering Applications of Artificial Intelligence*, *Expert Systems with Applications*, *Future Generation Computer Systems*, *IEEE Internet of Things Journal*, *IEEE Sensors Journal*, *IEEE Systems Journal*, *IEEE Transactions on Cybernetics*, *IEEE Transactions on Industrial Informatics*, *IEEE Transactions on Knowledge and Data Engineering*, *IEEE Transactions On Neural Networks and Learning Systems*, *Information Sciences*, *Intelligent Data Analysis*, *International Journal of Machine Learning and Cybernetics*, *International Journal of Medical Informatics*, *Internet of*

Things, *Journal of Artificial Intelligence and Soft Computing Research*, *Journal of Network and Computer Applications*, *Knowledge-Based Systems*, *MDPI Applied Sciences*, *MDPI Mathematics*, *MDPI Sensors*, *Neural Networks*, *Neurocomputing*, *Springer Applied Intelligence*, *Springer Cluster Computing*, and *Springer Machine Learning*. This result demonstrates that the research was conducted in reputable journals.

In addition, the distribution of studies across journals indicates that CD research is published in diverse scientific areas, with significant concentrations in the journals “Expert Systems with Applications” and “Information Sciences,” which have the highest number of publications, five and four, respectively. The selected journals also show an increasing

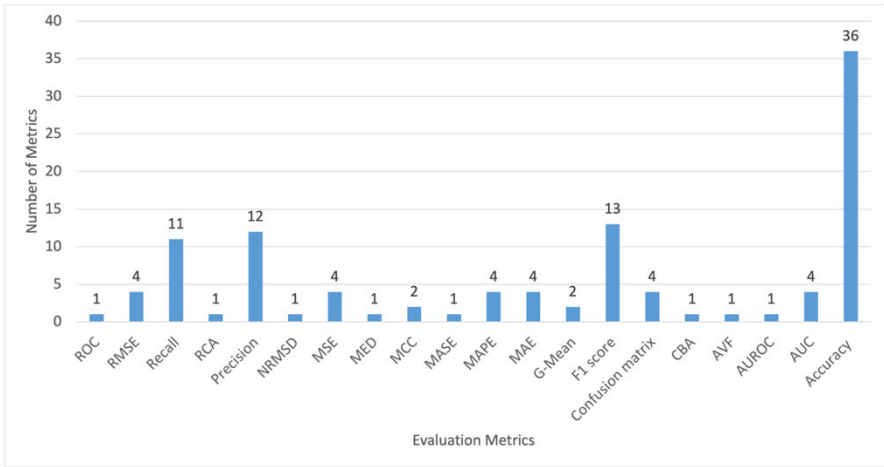


FIGURE 8. Characteristics of studies based on evaluation metrics.

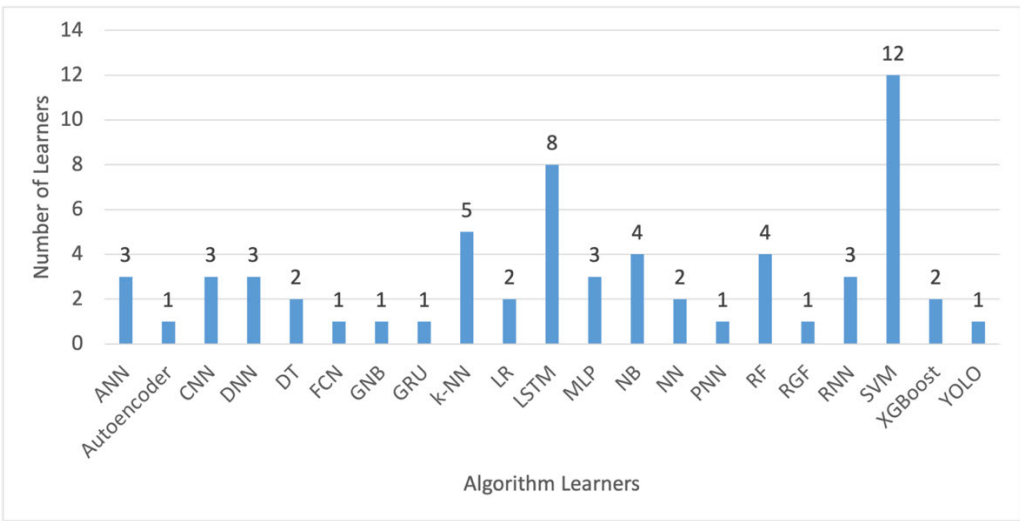


FIGURE 9. Characteristics of studies using learning algorithms.

emphasize on the practical deployments and real-world challenges of CD, in which the predictive accuracy of ML models suffers owing to unseen data shifts, particularly in time-sensitive applications such as IoT, industrial systems, and real-time decision-making. These results indicate that complications associated with CD detection have been reported in several studies.

Figure 7 shows the distribution of the scope of the selected study-based problems for ML. Most studies focused on classification tasks (60%), followed by time-series (12%), supervised (7%), imbalance (7%), unsupervised (6%), regression (6%), and semi-supervised (2%). This indicates that researchers are interested in scoping ML problems using classification methods. This indicates that most advancements in CD detection and adaptation can be attributed to the focus on classification problems within time-series contexts. In the future, distribution patterns could be improved by expanding the detection of CD in different ML contexts.

Figure 8 presents the attributes of the selected studies based on the metrics used for model evaluation. The evaluation metrics were accuracy, area under the curve (AUC), receiver operating characteristic curve (AUROC), architectural vulnerability factor (AVF), cost-benefit analysis (CBA), confusion matrix, f1 score, geometric mean (G-Mean), mean absolute error (MAE), mean absolute percentage error (MAPE), mean absolute scaled error (MASE), Matthews correlation coefficient (MCC), median absolute error (MED), mean squared error (MSE), normalized root-mean-square deviation (NRMSD), precision, root cause analysis (RCA), recall, root mean square error (RMSE), and receiver operator characteristic (ROC). The findings demonstrated that most studies employed accuracy as the primary measure to evaluate their models, whereas other studies concentrated on a range of alternative metrics to evaluate CD detection models. The accuracy evaluation metric can serve as the primary means for ML models because of its easy interpretation and direct

TABLE 5. A brief overview of studies categorized based on classification learning algorithms.

Classification Learning Algorithms	Year	Datasets	Description
LSTM [46]	2024	419 sensors	An AI-driven computing architecture for smart city applications in edge micro-data centers in a hybrid cloud-edge environment is proposed in this paper. A feedback-based method for automated CD detection and adaptation using LSTM algorithms and classification-based statistical tests based on classification is also presented.
LSTM [47]	2024	Power supply and posture	This study introduces QuadCDD, a novel CD classification framework that determines the drift length in data streams. In addition, it provides an analysis that helps make informed decisions and adjust to drift scenarios, ensuring consistent and reliable model performance in drifting data streams.
XGB, RNN [9]	2023	Electronic health records (EHR)	Data drift and its effects on sepsis prediction were examined to perform ML. This study defined data drift in sepsis and similar disease prediction. This approach could improve hospital patient monitoring systems that categorize dynamic disease risk.
MLP, GNB [48]	2023	Airlines, covtype, electricity	This paper introduces a chunk-based AWAE algorithm for nonstationary data stream classification in CD. The proposed AWAE dynamically adjusts its classifiers in response to concept changes.
MLP [49]	2022	Network and APT C&C and web categorization	This study uses the Nearest Class Mean (NCM) classifier to train a drift detector that effectively identifies the drift classes. In addition, existing data distribution detection and model retraining methods are not robust to attacker-induced drift or delayed attack detection.
k-NN, SVM [50]	2022	Image, Speech	This article presents TML-CD which is a CD solution that uses deep learning (DL) feature extractors and k-NN classifiers. The proposed solution also includes a hybrid adaptation module to handle CD during data generation.
YOLO [51]	2021	NTUST_EE Lobby	Human involvement in maintaining and updating analytical models is required to adapt to CD. This study addresses this issue and follows the trend of anthropomorphic ML by incorporating self-regulated learning from human pedagogy.
NB [52]	2020	Enron dataset	This study aims to enhance spam filtering by presenting a technique to identify CDs in email content. In addition, instances are classified as spam or non-spam using the KRR (KELM) based classification model.
RF [53]	2020	Electricity	This paper presents the Random Forest with Features Importance (RFFI), which is a novel algorithm for classifying data streams.
CNN, LSTM [54]	2019	CovtypeNorm, electricity, poker-lsn	This study classifies data streams with dynamic parameters, requiring model adaptation to changing patterns using CNN and LSTM models. The results also highlight the problem of window-based online learning rather than single-pass processing.
SVM [55]	2015	Moving hyperplane, letter forest, cover type	The authors introduce UELM-MapReduce, a parallel ensemble classifier based on ELM and MapReduce, to manage uncertain data streams. The uncertain data streams are sequentially trained into training chunks to construct a powerful parallel ELM-based ensemble classifier.
DT [56]	2013	NSL-KDD, soybean, image segmentation	This study presents an adaptive ensemble method for CD management and data stream class discovery. In addition, traditional classifiers in an ensemble are updated automatically to reflect changes in the concepts.

reflection of the model's predictive performance. In CD scenarios, maintaining the stability of the ML model in changing data distributions is essential. Thus, accuracy serves as a direct measure of a model's ability to adapt to new patterns. Future studies could benefit from deeper metric adoption to improve the robustness of CD detection models under various drift conditions and their adaptability to real-world applications.

Figure 9 presents the characteristics of the selected studies that investigated the implementation of different CD algorithms. The algorithms are an artificial neural network (ANN), autoencoder, convolutional neural network (CNN), deep neural network (DNN), decision tree (DT), fully convolutional network (FCN), Gaussian naïve Bayes (GNB), gated recurrent unit (GRU), k-nearest neighbors (k-NNs), logistic regression (LR), long short-term memory (LSTM), multilayer perceptron (MLP), naive Bayes (NB), neural networks (NN), perceptron neural network (PNN), random forest (RF), regularized greedy forests (RGF), recurrent neural network (RNN), support vector machine (SVM), eXtreme gradient boosting (XGBoost), and you only look

once (YOLO). The research findings tended to explore the use of SVM-based learning algorithms for CD detection. In addition, most studies have employed SVM and LSTM models for CD detection and adaptation because of their ability to handle dynamic data shifts. SVMs perform well in high-dimensional spaces and are useful for detecting shifts in various data distributions that are important in CD scenarios. Specifically designed for sequential data, LSTMs can capture temporal dependencies and adapt to gradual changes over time, which is a common feature of CD in time-series data. The findings indicate that future frameworks for CD detection and adaptation can incorporate advanced ML models, particularly for complex time-series data, as computational efficiency increases.

B. CLASSIFICATION OF SELECTED STUDIES

1) SUMMARY OF STUDIES BASED ON CLASSIFICATION AND REGRESSION ALGORITHMS OF LEARNERS

The classification of studies based on learners' CD CD-detection algorithms is presented in Table 5. The table also includes datasets, descriptions of their work,

TABLE 6. A brief overview of studies categorized based on regression learning algorithms.

Regression Learning Algorithms	Year	Datasets	Description
ANN [57]	2023	Static dataset from a real-world system	The authors present a technique for achieving a suitable balance between two requirements in a data stream using an ANN. In addition, the method efficiently updates the neural regression models using only the most recent measurement samples and a limited amount of buffered data storage space.
LR [8]	2022	UCI Electricity	This study introduces a method for addressing CD that uses profiles and a linear regression (LR) model, eliminating the need for costly retraining. Furthermore, using the profiles, the energy time-series prediction can effectively manage changes in the patterns over time.
RT [58]	2021	Google cluster trace dataset	The authors developed a predictive model, specifically utilizing regression trees (RT) to ensure the applicability of their experimental findings. The study employed this model to predict disc failures using the Backblaze disc stats dataset and job failures using the Google cluster trace dataset.
SVR [59]	2020	Crime and Sensors dataset	This paper introduces a framework to adapt trained models to changes in the CD of the underlying query workload distribution. The study also focuses on lightweight online statistical learning models, specifically, SVR models, which are used to estimate analytical query answers efficiently and accurately.
MSR [60]	2017	Synthetic and real-world datasets	This study presents a method for incorporating CD detection into covariate shift adaptation using multi-stream regression (MSR) and asynchronous processing. Furthermore, the proposed technique was empirically tested on synthetic and real-world datasets.

publication years, and algorithm learners. Classification-based algorithm learners play a crucial role in detecting, handling, and serving CD adaptors. Selected learners with complementary characteristics, such as SVM, ANN, and DT, enhance drift detection capabilities. Algorithm learners are essential components of CD detection in time-series data and data streams. In addition, their ability to sense changes and adapt accordingly is critical for maintaining the accuracy and reliability of the model in dynamic environments. According to a summary of previous studies, SVM, LSTM, k-NN, and NB algorithms are the most frequently used to address CD-related problems. In addition, previous studies have combined various learner methods to compare their performance, leading to the development of intelligent hybrid models. Table 6 summarizes the classification of studies based on regression algorithm learners for CD detection. The table also includes datasets, descriptions of their work, publication years, and regression algorithm learners. Regression-based algorithm learners play a crucial role in detecting, handling, and interpreting CD by predicting continuous values. Learners with complementary characteristics, such as ANN, LR, RT, SVR, and MSR, enhance drift detection capabilities. However, regression algorithm learners are essential components of CD detection in time-series data and data streams. In addition, their ability to sense changes and adapt accordingly is critical for maintaining the accuracy and reliability of the model in dynamic environments.

2) SUMMARY OF STUDIES CLASSIFIED BASED ON MODEL PERFORMANCE

The studies classified according to the model performance for CD drift detection and adaptation in a general context are listed in Table 7. Most previous studies have focused on data streaming, imbalances, and anomaly detection. In contrast, CD detection is based on the evaluation of model performance metrics. For example, abrupt increases

in error rates, such as the mean squared error (MSE), classification error, or abrupt decreases in confidence scores, indicate drift. Various metrics, such as accuracy, precision, recall, and F1 measure, reveal drift characteristics, such as changes in specific classes or larger changes in the distribution. Variations in feature importance help identify the affected features, facilitating root cause analysis. These metrics also assess the efficacy of adaptation strategies, such as re-training and online learning, by comparing the performance before and after drift. Selecting appropriate metrics specifically designed for a given application and data is critical. Performance metrics serve as tools for identifying and addressing CD proactively. Additionally, the performance of the model was evaluated using appropriate evaluation metrics after training the ML models on various CD detection and adaptation datasets. For example, the authors who investigated class imbalance employed ACC, REC, PRE, and F1 to evaluate their predictive models to ensure that the models were effective in significant instances within imbalanced datasets. Metrics such as MSE and MAE are employed in time-series forecasting because of their accurate measurement of error magnitude over time. Finally, the targeted use of evaluation metrics highlights model performance and enables meaningful comparisons between studies.

3) SUMMARY OF STUDIES CLASSIFIED BASED ON A TECHNIQUE THAT HANDLES CONCEPT DRIFT

The studies classified according to the techniques used to handle the CD types are presented in Table 8. Most previous studies used the ADWIN, HDDM, and DDM techniques to overcome drift problems. Common drift types are real, incremental, gradual, sudden, and recurrent drift. However, CD can surface in several ways, including gradual shifts and sudden changes. Each of these presents specific challenges for detection and adaptation. The summarized

TABLE 7. A brief overview of studies categorized by model performance for CD detection and adaptation in a general context.

Performance Focus	Model Performance								
	ACC (%)	REC (%)	PRE (%)	F1 (%)	AvFβ (%)	CBA (%)	RMSE	MSE	MAE
Data stream [61]	0.98	-	1.00	0.99	-	-	-	-	-
Imbalance data [62]	-	-	-	-	87.35	81.04	-	-	-
Data stream [63]	-	-	-	-	-	-	0.82	1.09	0.53
Domain adaptation [64]	89.07	-	-	-	-	-	-	-	-
Drift adaptation [65]	90.17	87.03	89.68	88.34	-	-	-	-	-
Drifting data points [66]	0.66	-	-	-	-	-	-	-	-
Data stream [67]	96.1	96	96	96	-	-	-	-	-
Data stream [68]	98.5	-	-	0.85	-	-	-	-	-
Class imbalance [69]	89.53	74.03	84.21	78.70	-	-	-	-	-
Anomaly detection [70]	93.52	94.30	91.84	92.9	-	-	-	-	-
Time-series [71]	-	-	-	-	-	-	-	0.042	0.166
Data stream [72]	63.65	-	-	-	-	-	-	-	-
Data stream [73]	93.0	0.94	96.0	94.0	-	-	-	-	-
Dynamic logs [74]	-	95.2	98.2	96.7	-	-	-	-	-
Decaying rate [75]	99.35	-	-	-	-	-	-	-	-
Data stream [76]	95.44	-	-	-	-	-	-	-	-
Data stream [77]	63.64	-	-	-	-	-	-	-	-
Data stream [78]	0.66	-	-	-	-	-	-	-	-
Time-series [18]	-	-	-	-	-	-	-	6.63	2.25
Data stream [79]	-	-	-	75.63	-	-	-	-	-
Data stream [80]	97.06	-	-	-	-	-	-	-	-
Time-series [81]	-	-	-	-	-	-	24.76	-	-
Time-series [82]	83.78	-	-	-	-	-	-	-	-
Data stream [6]	0.80	-	-	-	-	-	-	-	-
Data stream [83]	86.37	-	-	-	-	-	-	-	-
Data stream [84]	-	0.65	0.54	-	-	-	-	-	-
Pattern recognition [85]	96.02	-	-	-	-	-	-	-	-

study technique combines the years and types of CD to provide a more in-depth understanding of the performance of algorithmic techniques for CD detection. In addition, each technique employed for the detection of various types of CD exhibits special qualities suited to the type of drift pattern. For example, ADWIN and KSWIN dynamically change with changes, which qualifies them as either gradual or sudden shifts in data distribution. CUSUM and EWMA use cumulative changes or weighted averages to facilitate the rapid identification of incremental drifts. EDDM and HDDM use statistical distance measures that effectively detect gradual drifts by tracking slow changes. Finally, this can enable future research to address the obstacles posed by CD types in real-world applications across multiple domains by selecting appropriate algorithms.

V. MAJOR FINDINGS BASED ON THE RESEARCH QUESTION

A. RQ1: WHICH ENSEMBLE-BASED LEARNING METHODS ARE MOST EFFECTIVE FOR DETECTING CD IN TIME-SERIES DATA USING CLASSIFICATION AND REGRESSION APPROACHES?

To identify existing ensemble-based learning approaches for CD detection in time-series data (RQ1), this study confirmed

the existence of several approaches (Figure 10). In addition, this question reveals how their characteristics affect detection accuracy and adaptability to other methods. CD poses a significant challenge in time-series data analysis, rendering models ineffective as underlying patterns change. Ensemble-based learning methods have emerged as promising solutions because of their inherent diversity and robustness. Investigating their efficacy and comparing them with single-classifier approaches is critical for making informed decisions.

Several ensemble-based techniques have shown potential for detecting CD in time-series data. For example, Thalor and Patil [100] proposed an ensemble-based classification algorithm for nonstationary data streams (ENSDDS) with an emphasis on two-class problems for the incremental learning of nonstationary data streams. The authors also found that ENSDDS outperformed Learn++. NSE in various evaluation metrics. To address the problem of pattern drifting in ML applications under nonstationary operating conditions, Liu and Zio [101] presented an ensemble model based on a support vector regression (SVR) model with streaming data points. In addition, the proposed method reduces computational complexity by updating a portion of the newly available data and enabling the timely tracking of ongoing patterns using feature vector selection. According

TABLE 8. An overview of studies is categorized based on a technique that handles concept drift.

Technique	Year	The type of drift				
		Real	Incremental	Gradual	Sudden	Recurrent
ADWIN, KSWIN [46]	2024	✓				
QuadCDD [47]	2024	✓	✓	✓		✓
CUSUM, EWMA, PHT [63]	2023		✓			
HDDDM, CDBD [61]	2022	✓	✓	✓	✓	
ADWIN [18]	2022			✓		
ADWIN, SAM [50]	2022			✓		
ADWIN, EDDM, KSWIN [65]	2022			✓	✓	
ADD [68]	2022		✓	✓	✓	
RDCAL [69]	2022		✓			
ADWIN [8]	2022		✓		✓	
EDDM, HDDM, GMA [79]	2022		✓	✓	✓	
GGM [86]	2022			✓		
MDDT, CUSUM [82]	2021	✓	✓			
DDM, HDDM [87]	2021		✓			✓
EDDM, DDM, ADWIN [72]	2021		✓			
HDWM [88]	2020			✓	✓	
ACDDM [89]	2020		✓	✓	✓	
CSDD [90]	2019			✓	✓	
FSDD, FPDD [91]	2018	✓		✓		
FTRL-ADP [75]	2018	✓		✓		
WSTD [84]	2018	✓		✓		
AL-ELM [85]	2017		✓			
MOS-ELM [92]	2016			✓	✓	✓
NDE [93]	2016			✓	✓	✓
FHDDM [94]	2016	✓	✓	✓		
DDM-PHT [95]	2015				✓	
DOED [76]	2015			✓	✓	
HDDM [96]	2015		✓	✓		
OAUE [77]	2014		✓	✓	✓	
AGE [97]	2014			✓	✓	
ACCD [78]	2014			✓		
ADDM [98]	2014			✓	✓	
Learn++.CDS [99]	2013		✓	✓	✓	

to Cavalcante and Oliveira [4], addressing CD in financial time-series predictions improves forecasting accuracy. The authors employed an ensemble Extreme Learning

Machine (ELM) with explicit drift detection and updated the decision model online when a CD occurred in the data. This study found that drift detection can improve the

TABLE 9. Summary of ensemble-based learning concept drift detection techniques.

Models	Characteristics
ENSDS [100]	Ensures that it is resistant to drift by generating new classifiers as the data changes and assigning weights based on recent prediction performances.
SVR [101]	Retrained or incrementally updated, rendering it suitable for real-time drift adaptation in time-series contexts, such as financial forecasting, where data distributions frequently change.
ELM [4]	- It can be used as part of an ensemble to capture shifting patterns in time-series data. - Providing nonlinear activation functions allows capturing complex temporal dynamics, making it useful in environments with frequent drift.
OS-ELM [102]	It provides a specific design for adaptive learning in nonstationary environments, thereby facilitating CD detection in time-series data.
DEMS [103]	- Identifying groups of similar classifiers within an ensemble based on performance patterns and selecting representatives from each cluster. - Enables the ensemble learners to adjust to variations in data distributions by maintaining a range of classifiers adapted for different distributional conditions.
DES [104]	An approach to detect CD in time-series data enhances the adaptability of the ensemble by selecting the most relevant classifiers from a pool based on their local performance in the current data context.

prediction time of online sequential ELM while maintaining accuracy.

A similar study by Grim and Gradwohl [102] introduced an online sequential extreme learning machine (OS-ELM) to address the challenge of forecasting data streams using CDs. The proposed method involves the creation of three ensembles that use the OS-ELM and employ high-performance techniques. High-performance techniques involve advanced computational approaches that improve the speed and efficiency of ensembles for regression and time-series forecasting using OS-ELM. The findings showed that high-performance ensembles outperformed the corresponding serial versions. Additionally, drift-based ensemble member selection using clustering (DEMSC) has been proposed for adaptive selection and a combination of time-series forecasting models. This assumption facilitates the dynamic selection of initial ensemble-based model candidates by implementing a performance drift detection mechanism. Additionally, a comprehensive empirical analysis of the proposed method was conducted to assess both the standardization error and scalability of the proposed approach using time-series data from various real-world domains [103]. Boulegane et al. [104] reported that dynamic ensemble selection (DES) is a promising method for forecasting time-series in real time using multi-target regression (MTR). The authors introduced a new method to use potential interconnections within the behavioral characteristics of base models using incremental MTR to achieve DES and enhance model performance.

Furthermore, several characteristics of ensemble-based learning approaches significantly affect their detection accuracy and adaptability to other classification methods, despite the CD in the time-series data. For example, using diversity to combine multiple base learners trained on different data samples or algorithms, ensembles can inherently avoid individual model biases and capture a broader spectrum of

potential patterns. This leads to more robust and generalizable detections, especially when shifting concepts, which is difficult for individual models. In addition, the diversity of available models was used to improve the performance of each base learner with different parameter settings [104]. However, dynamic adaptation plays a vital role; unlike static single models, ensembles incorporate mechanisms, such as dynamic weighting, to prioritize base learners that perform well in the current concept. This enables faster adaptation to both gradual and abrupt drifts while maintaining pace with evolving data patterns. For example, [101] employed a dynamic ensemble selection strategy that was integrated into the methodology to select the sub-models that were most relevant to the new data point to derive the prediction while simultaneously minimizing the impact of irrelevant models. However, compared to other classification approaches, ensembles frequently demonstrate superior adaptability in dynamic environments because of their inherent diversity and dynamic update mechanisms. The characteristics of ensemble-based learning provide significant advantages in terms of detection accuracy and adaptability when detecting CD in time-series datasets. The techniques for detecting CD using ensemble-based learners (Table 9). Finally, it was found that ensemble-based learning with classification and regression approaches based on ENSDS, SVR, ELM, OS-ELM, DEMSC, and DES were the most commonly used methods for detecting CD in a time-series environment. This could be due to their inherent diversity and effectiveness in mitigating CD.

B. RQ2: WHAT ARE THE VARIOUS ML MODELS USED IN A GENERAL CONTEXT TO DETECT CD AND HOW DO THEY PERFORM IN TERMS OF ACTIONABILITY IN DRIFT DETECTION AND ADAPTATION TASKS?

Several ML models have been employed for CD detection in time-series forecasting tasks, where each learner exhibits

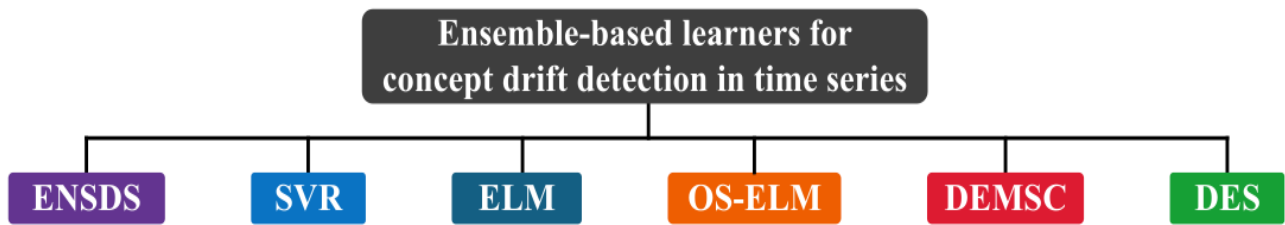


FIGURE 10. Ensemble-based learners for concept drift detection in time-series data.

benefits and drawbacks in terms of their actionability. Based on these findings, SVM, k-NN, and LSTM have been widely used because of their efficacy in drift detection. In addition, both ML models possess specific advantages in CD detection: the SVM and k-NN models are robust against feature-based changes, whereas LSTM captures temporal dependencies, making them appropriate options for various types of drifts in time-series applications. Learners can be classified into supervised, unsupervised, and DL models.

Supervised learners are crucial for CD detection and adaptation. For example, Tang et al. [64] used domain adaptation to decrease CD in incremental learning through a method called (incremental support vector machine with domain adaptation) which is based on SVM. The experimental results demonstrated a greater efficiency when tested using both real and synthetic data. Similar research by Jain et al. [70] employed SVM classifier to detect CD in network anomaly detection. The model was retrained using statistical tests. The authors focused on several types of CD, including sudden, gradual, and incremental drifts. In the following experiments on three datasets, namely, the Testbed Dataset, NSL-KDD, and CIDDs-2017, SVM demonstrated better classification accuracy. Hinder et al. [66] presented a novel supervised ML approach using RF, k-NN, and DT as general theoretical frameworks to address the drift localization problem. In addition, the authors provided a theoretical framework that relates probabilistic classification to drift localization. The empirical results demonstrate that the proposed method is a promising approach for both drift localization and additional feature relevance analysis.

In addition, unsupervised learning can be employed to address drift detection challenges. For example, Kamin-skyi et al. [61] proposed an autoencoder-based unsupervised drift detection algorithm for multivariate streaming data. The proposed approach is known as the autoencoder-based concept drift detector (AECDD), which is an unsupervised drift detection model based on reconstruction that tracks changes in data distribution using an autoencoder. AECDD was evaluated using synthetic data designed to assess the capacity of the model to capture contextual information, and the proposed approach demonstrated positive results. Huang et al. [85] introduced an innovative unsupervised adaptive approach that reduces the degradation of long-term myoelectric control at low computational cost. In addition,

the authors presented a universal incremental least-squares support vector classifier (LS-SVC) and particle adaptive classifier (PAC) that generated a particle adaptive learning strategy. Realistic data collected over a single day was used to validate the proposed method.

However, the rapid development of DL models such as LSTM, ANN, and RNN models offers effective drift detection capabilities. It was found that [46], [47], [63], [71], [80], [81], [105] used LSTM-based learners for CD detection, adapting, understanding, and analyzing CDs. Moreover, to address CD in streaming data, [6] used a deep neural network to propose a technique known as selective ensemble-based online adaptive (SEOA). The adaptive depth unit builds shallow features with deep features and adjusts the information flow of the neural network in response to nearby data variations to improve the convergence of the online DL model. The experimental results demonstrate that SEOA can combat CD and is generalized and robust. Table 10 presents various ML model techniques for detecting CD in a general context. Finally, the most frequently employed ML models for CD detection in a general context were investigated. According to these findings, SVM, k-NN, and LSTM are widely used owing to their efficacy in drift detection. The study also found that other deep neural network models, such as SEOA, can effectively combat CD in a general context.

C. RQ3: WHAT ALGORITHMS HAVE BEEN USED IN THE LAST DECADES FOR ANALYSING AND HANDLING CD IN TIME-SERIES DATA AND DATA STREAMS?

The paper found that Several algorithms have been proposed to address this problem. For example, DDM, ECDD, and ADWIN are available for handling and analyzing CD [4], [72]. Wang et al. [82] employed a multiscale drift detection test (MDDT) algorithm to identify CD in a nonstationary time-series environment. The authors efficiently used an archive to identify sudden changes in feature values that required immediate adaptation. The results demonstrate that the proposed method outperforms other methods in handling abrupt shifts and achieves the highest recall score for identifying drift points. In a similar study conducted by Ramanan et al. [106], the Unsupervised Temporal Drift Detector (UTDD) algorithm was employed to identify real-time CDs in time-series data. The goal of this study was to detect seasonal variation and temporal CD in time-series

data without ground truth and then adjust the ML models to handle CD to improve their generalizability.

In addition, Dehghan et al. [93] employed an NDE algorithm that depended on ensemble classifiers to detect CD. The authors' method analyzes samples individually and tracks the distribution of the ensemble error to identify potential drifts. The experimental results demonstrate that the proposed method can effectively detect and adapt to several types of CDs and outperforms existing methods. Pesaranghader and Viktor [94] introduced the FHDDM method as a means of detecting drift points in data streams. This was achieved using a sliding window and Hoeffding inequality. The experimental results verified that the FHDDM had a lower detection delay, fewer false positives, and fewer false negatives when drifts were detected. Table 11 summarizes the existing algorithms used over the last decade to detect, handle, and analyze CD in time-series and data streams. Finally, it was discovered that various algorithms, such as ADWIN, HDDM, and ADD, have been used to detect, handle, and analyze CD in time-series and data streams.

D. RQ4: WHAT ARE THE MAIN STEPS TO ADDRESS CONCEPT DRIFT PROBLEMS, SPECIFICALLY IN TIME-SERIES DATA AND THE GENERAL CONTEXT?

This question aims to identify a general pipeline for detecting CD in time-series data and general contexts, which involves several key steps that ensure that the model remains accurate and reliable despite changes in the underlying data distribution. In addition, to address CD in time-series data, it is necessary to understand the nature of the data and the detection, characterization, adaptation, and evaluation of model performance to ensure that its accuracy is maintained. The basic steps for addressing CD are the same for both general contexts, with time-series data provided in Sections I and II.

1) ADDRESSING CONCEPT DRIFT IN THE GENERAL CONTEXT

- (a) **Step 1. Nature of data:** The nature of the data included no sequential dependency, and the data were independent and identically distributed. The data types included tabular data, text, and images without inherent temporal order. For example, a dataset based on text data streams was validated using the proposed model for three-layer CD detection. These datasets contained drift categories [108].
- (b) **Step 2. Concept drift detection:** Detection CD involves identifying drift using statistical tests, monitoring prediction errors, or specialized algorithms. However, the same algorithms, such as Page-Hinkley, CUSUM, and EDDM, can be applied in different contexts. For example, the EDDM, DDM, and ADWIN methods have been thoroughly analyzed to investigate the CD problem in fake-review detection. The proposed method was also tested using the Yelp consumer electronics dataset from Yelp.com [72].

- (c) **Step 3. Concept drift characterization:** In general, CD characterization involves determining the type of drift (sudden, gradual, incremental, or recurring) to select suitable mitigation methods for any scenario. In [109], the authors characterized their automated method for detecting CD based on sudden and gradual drifts in business processes using execution traces. Additionally, an analysis of synthetic logs revealed that the proposed approach effectively identified typical process changes and nested compositions, surpassing state-of-the-art baselines for sudden and gradual drifts.
- (d) **Step 4. Model adaptation:** Adaptation of CD in a general context also involves re-training models using recent data, incrementally updating them, employing ensemble methods, or using techniques such as weighted re-training. For example, an optimized DL and adaptive sliding window approach were employed for the CD adaptation framework. The authors also analyzed dynamic streaming data based on improved DL techniques and adaptive and sliding window methods, which work well with limited memory and time [110].
- (e) **Step 5. Evaluation and validation:** The adapted models were continuously evaluated and validated by employing performance metrics, such as ACC, ROC, and AUC, as well as cross-validation techniques, to ensure the accurate capture of new data patterns. In [80] and [111], the authors used ACC, ROC, and AUC to evaluate their models for CD detection, adaptation to fake news, and cyberattack scenarios.

2) ADDRESSING CONCEPT DRIFT IN TIME-SERIES DATA

- (a) **Step 1. Nature of data:** Time-series data consist of a sequential collection of data points that are gathered and combined at equally spaced intervals. For example, the sequential dependency in time-series data comprises sequential observations taken over time, in which each data point relies on the previous point. Temporal trends frequently exhibit trends, seasonality, and autocorrelations. In [112], the authors employed two public real-world time-series datasets to evaluate their proposed CD detection method based on incremental learning in nonstationary environments.
- (b) **Step 2. Concept drift detection:** CD detection in time-series data can be accomplished via statistical tests or by employing drift detection algorithms such as Page-Hinkley, CUSUM, DDM, and ECDD. The detection method focuses on the temporal sequence of data points. For example, the DDM and ECDD methods have been employed to detect CD in financial time-series predictions to enhance the prediction accuracy of the proposed model [4].
- (c) **Step 3. Concept drift characterization:** The characterization of time-series data involves identifying temporal patterns, such as seasonal fluctuations or

TABLE 10. Summary of machine learning techniques for detecting concept drift in a general context.

Models	Characteristics
SVM [64]	• SVM models can detect CD because the model can handle high-dimensional and nonlinear patterns. • It also provides flexibility in handling various kernel functions, which improves its ability to identify complex relationships in time-series data and detect drift.
k-NN [66]	• A non-parametric method for detecting CD in time-series data that uses continuous evaluation of neighborhood changes around a data point. • Respond to shifts in data distribution by updating the reference set with new instances, thereby enabling drift.
LSTM [46]	• Retaining long-term dependencies enables one to capture temporal patterns and adjust to distributional changes over time. • Monitor changes in sequence prediction errors; rising errors indicate changes in the underlying data distribution.
AECDD [61]	• Ability to learn low-dimensional representations of input data and recognize deviations from previously learned patterns. • Reduce the reconstruction error for normal patterns by compressing data into latent space before reconstructing the patterns, thereby making them sensitive to anomalies that indicate potential CD.

TABLE 11. Summary of algorithms used in the past decade to handle concept drift.

Authors	Year	Algorithm	Data Focused	
			Time-series Data	Data Stream
Mehmood et al. [46]	2024	ADWIN, KSWIN	✓	✓
Alencar et al. [63]	2023	CUSUM, EWMA, PHT	✓	✓
Heidrich et al. [8]	2022	ADWIN	✓	
Ramanan et al. [106]	2021	UTDD	✓	
Priya et al. [52]	2020	GA		✓
Fields et al. [105]	2019	ML (MLP, LSTM, CNN, GRU)	✓	
Cabral et al. [91]	2018	FSDD, FPDD		✓
Xu et al. [107]	2017	AL-ELM	✓	✓
Mirza et al. [92]	2016	MOS-ELM		✓
Cavalcante et al. [4]	2015	DDM, ECDD	✓	✓
Brzezinski et al. [77]	2014	OAUE		✓
Ditzler et al. [99]	2013	Learn++.CDS	✓	

long-term tendencies that can influence drift. In addition, CD can be used to determine whether the drift is sudden, gradual, incremental, or recurring to determine the most effective mitigation strategy. A recent study investigated a new method for detecting CD in nonstationary environments during incremental learning. Furthermore, the authors characterized different types of drift: sudden, gradual, and incremental [112].

- (d) **Step 4. Model adaptation:** The adaptation technique in time-series models involves re-training recent data, incrementally updating models, or employing ensemble methods. Methods such as sliding windows and weighted re-training are highly efficient. To address the challenges posed by sudden or consistent changes in normal behavior, an ensemble DL model with CD adaptation was employed for time-series anomaly detection. The model is trained incrementally as new

data become available and can adapt to changes in data distribution [113].

- (e) **Step 5. Evaluation and validation:** Regular evaluation and validation of the adapted time-series models require specific validation methods, such as cross-validation techniques, and performance metrics to ensure accuracy, such as ACC, MAE, and RMSE. For example, RMSE and MASE evaluation metrics have been used to evaluate the CD model in a time-series context [8], [81].

VI. TAXONOMY

This section presents a taxonomy for the AI learner roadmap, which provides a brief overview of advances in handling, detecting, and adapting to CD. However, understanding the different types of CD is essential for the development of robust AI systems. Recognizing a specific drift type and its characteristics enables researchers to develop appropriate

detection methods that ensure the effectiveness of models in dynamic environments. The taxonomy was developed after a critical review of existing literature and the present experimental evaluation of the proposed roadmap for CD detection. Subsection B provides an evaluation of AI models for CD detection in time-series applications.

A. ROADMAP FOR CONCEPT DRIFT DETECTION

An in-depth analysis of the ability of AI-based learners to detect CD based on categories such as speed and severity. The methods were classified according to their speed and severity, which were categorized as incremental, gradual, sudden, or recurring (Figure 11).

1) AI-BASED LEARNERS FOR DETECTING INCREMENTAL DRIFT

Incremental CD occurs when there is a minimal and continuous change in the original data distribution and the relationship between features and target variables. The effects of incremental changes in CD occur in small, gradual increments over time, making them difficult to detect in real-time. AI-based learners can also detect this type of CD. For example, Wang et al. [47] presented a quadruple-based approach for understanding CD in data streams (QuadCDD) framework, which is a novel deep neural network (DNN)-based learner that detects and predicts CD, but also predicts incremental drift. The quadruple provides a comprehensive picture of CD, including its start, end, severity, and type. In addition, recent studies have proposed robust models for detecting incremental CD using SVM, GNB, and MLP [48], [83].

2) AI-BASED LEARNERS FOR DETECTING GRADUAL DRIFT

Gradual CD occurs when there are gradual and noticeable changes in the target data distribution. The effects of gradual CD include long-term effects. However, this approach allows for detection earlier than incremental drift. Disabato and Roveri [50] presented a Tiny Machine Learning (TML) solution for CD that integrates k-NN, SVM, and NN learners. These learners include a hybrid adaptation module that can handle the gradual drift that affects the data-generation process. The accuracy of the model in the speech-command-identification scenario was evaluated over time with gradual and abrupt noise addition. In addition, a framework for classifying CD based on FCN learning was proposed. This framework involves offline pre-training of a model on data streams with known drifts and then fine-tuning the model online to improve detection accuracy [68].

3) AI-BASED LEARNERS FOR DETECTING SUDDEN DRIFT

A sudden CD can be interpreted as an abrupt and significant change in the original data distribution at a particular time point and as a relationship between the features and target variables. The effect of a sudden CD leads to abrupt and unforeseen behaviors, which presents a significant challenge to models trained on previous data. AI learners have been

analyzed to detect sudden drift [9]. This study also introduced XGBoost and RNN learners to model the impact of data drift on model performance, such as sudden and potentially catastrophic events, such as COVID-19. In addition, this study contributes to the creation of better patient monitoring systems in hospitals that can stratify patients according to their risk of changing their disease states. A novel chunk-based technique for classifying nonstationary data streams is proposed, and GBN and MLP are used to detect sudden drifts. To respond effectively to CD, the classifier ensemble technique employs a shifting ensemble line-up [48].

4) AI-BASED LEARNERS FOR DETECTING RECURRENT DRIFT
Recurrent CD signifies a situation in which an old concept reappears after absence. However, this was similar to gradual CD. The effects of recurrent CD include changes in previously observed concepts. Sun et al. [71] addressed recurrent drift in process industry streaming data by employing DLSTM, LSTM, and OAR learners. The proposed predictive method supports the handling of streaming data by using hybrid recurring CDs in the processing industry. In addition, the prediction task for recurring concepts is distributed across multiple independent sub-models. Further investigations presented a novel algorithm learner for classifying data streams using a random forest, where the feature importance metric was used as the drift detector [53]. Furthermore, the proposed algorithm combines the ability of ensemble methods to handle slow changes in data streams with a novel method for detecting CD occurrences.

B. EXPERIMENTAL EVALUATION

The evaluation of AI models for CD detection using time-series data involves several experimental methods. This study presents an experimental evaluation of the CD roadmap, which includes data evaluation, performance metrics, evaluation procedures, and baseline methods.

1) EVALUATION DATA

The data for CD detection using AI models will be evaluated using real-world time-series data such as financial, weather, and sensor data. These datasets ensure that the detection algorithms can efficiently handle the complex nature of real-world drifts after allowing model performance evaluation in an authentic scenario. In addition, the use of real-world data helps to confirm the effectiveness of the model in addressing real-world challenges associated with CD in dynamic environments.

2) PERFORMANCE METRICS

Performance metrics are crucial for determining the effectiveness of the proposed roadmap for CD detection in time-series environments. Performance evaluation metrics were employed based on classification and regression problems. Collectively, these metrics ensure a robust evaluation framework for AI models that address both the classification and regression aspects of CD detection. In addition, the accuracy,

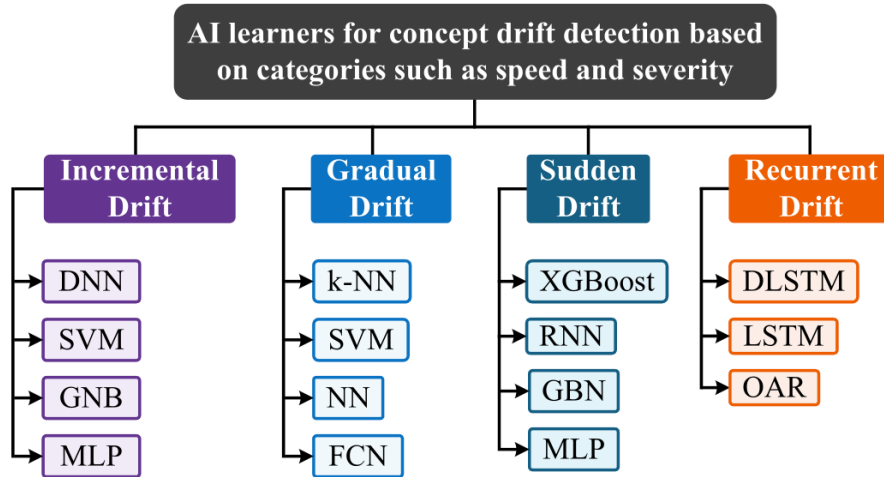


FIGURE 11. Illustration of AI learners for concept drift detection.

precision, recall, F1 score, and AUC were based on the classification metrics. RMSE, MSE, and MAE are regression metrics. The definitions and mathematical equations for the evaluation metrics are as follows:

- (a) **Accuracy (ACC):** ACC measures the proportion of correctly identified instances (both drift and no drift) among the total number of instances.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

- (b) **Precision (P):** This measures the proportion of the positive predictive value and indicates the proportion of detected drift, which is the actual drift of the model.

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

- (c) **Recall (R):** This measures the proportion of actual drift detected by the model.

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

- (d) **F1 score (F1):** The F1 score measures precision and recall, providing a balance between the two metrics.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (4)$$

- (e) **Root mean square error (RMSE):** RMSE measures the square root of the average squared difference between the predicted and actual values.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2} \quad (5)$$

- (f) **Mean squared error (MSE):** This represents the average of the squared differences between the predicted and actual values.

$$MSE = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2 \quad (6)$$

- (g) **Mean absolute error (MAE):** This represents the absolute difference between the predicted and actual values.

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i| \quad (7)$$

- (h) **Area under the curve (AUC):** AUC measures the ability of the model to distinguish between drift and no-drift cases across different threshold settings. The area under the receiver operating characteristic curve was also calculated.

3) EVALUATION PROCEDURE

Extensive evaluation procedures are required to ensure accurate results when evaluating the proposed CD roadmap for time-series data. The predictive evaluation, holdout method, and cross-validation evaluation procedures are described as follows:

- (a) **Prequential evaluation:** This method involves constant updating and testing of the model as new data arrives. Each new data point was used to test the model, followed by training. This method also simulates real-world scenarios in which the data streams are constant, enabling dynamic evaluation of the model's ability to detect CD over time.
- (b) **Holdout method:** The data were split into training and testing sets. The initial training set was used to train the model, which was then tested. This standard approach improves the model performance on previously unseen data and is useful for the preliminary benchmarking of drift detection capabilities.
- (c) **Cross validation:** Model robustness is guaranteed using cross-validation methods, including k-fold cross-validation. The data were then split into k subsets. Each time, the model was trained and validated by using a different subset as the validation set. This

approach provides a complete evaluation by averaging the performance over several data splits and helps understand the generalization capacity of the model in the presence of CD.

4) BASELINE METHODS

The baseline method compares the performance of AI learners with conventional methods for detecting CD in time-series data. However, standard drift detection techniques such as Page-Hinkley, ADWIN, DDM, and EDDM have been used as baselines. The performance of AI learners was evaluated using these established methods to ensure a thorough assessment of their effectiveness in detecting CDs.

C. COMPARATIVE ANALYSIS

This section compares the roadmap for CD detection with well-known baseline methods in the field by assessing their effectiveness and novelty. The roadmap uses advanced AI techniques such as SVM, GBN, NN, and XGBoost to detect various types of CD. These methods have several advantages over traditional approaches because of their ability to model complex, nonlinear relationships. The comparative analysis was divided into three main categories: detection capability, adaptability and flexibility, and performance and scalability.

1) DETECTION CAPABILITY

Well-known baseline techniques, such as DDM and EDDM, are effective at detecting abrupt and gradual drifts; however, their effectiveness is reduced in scenarios involving incremental or recurrent drifts because the technique relies on fixed statistical thresholds. In addition, their limitations include sensitivity to noise, false alarms, and difficulty in handling class-imbalance scenarios [114], [115]. Although the proposed DNN and XGBoost can capture complex patterns in the data, they are better at detecting incremental and recurrent drifts. These models can adaptively learn from and adjust to changes in data distribution, thereby providing a more sophisticated detection mechanism.

2) ADAPTABILITY AND FLEXIBILITY

Baseline techniques, such as ADWIN and DW_HDDM, are adaptable for handling several types of drifts. However, their performance can be reduced owing to the selection of parameters and the assumption of specific distribution properties. However, their limitations include computational complexity and detection delays [116], [117]. The adaptability of the proposed method to AI, including DNNs, XGBoost, and RF, allows it to dynamically adjust to evolving data streams. Their natural flexibility in handling various data distributions improves their robustness in real-world applications.

3) PERFORMANCE AND SCALABILITY

Conventional techniques centered around a baseline are computationally efficient and simple to implement; however,

many struggle to handle high-dimensional or complex time-series data because of scalability issues. For example, the WSTD method is limited because it is computationally expensive due to the sorting of observations [84]. Despite their computational complexity, AI techniques exhibit superior scalability when applied to large and high-dimensional datasets. The XGBoost and DNN techniques exhibit exceptional performance when handling large datasets.

VII. DISCUSSION

This section presents the findings concerning the RQs described in Section III. This study summarizes the key conclusions by analyzing and assessing empirical research on CD detection using time-series data. Because of the nonstationary characteristics of real-world data, CD detection is essential for reliable time-series analysis and forecasting. Over time, the underlying processes that generate data can evolve, rendering the historical patterns inapplicable to future predictions. By identifying these CDs, the efficacy of the model could be maintained. Early detection enables timely adaptation, such as re-training models with the new data distribution, which results in more reliable and generalizable forecasts. Additionally, time-series analyses and forecasting models are susceptible to CD, which can occur incrementally, gradually, suddenly, or recurring. Understanding these drifts is essential for creating robust forecasting models that can adapt to the evolving characteristics of real-world data.

Several methods are available for detecting CD in time-series data. Statistical techniques such as monitoring the average and variability can detect deviations from existing data distributions. The windowing method compares recent data with historical windows to identify changes in distribution patterns. This study focuses on ensemble-based learning techniques that are efficient in detecting CD in time-series data. These advantages allow multiple learners to create robust models with improved adaptability to changing data patterns. The core functionalities of ensemble-based learning include diverse base learners, error correction and aggregation, and continuous learning. In nonstationary data streams, an ensemble-based classification algorithm was used to detect CD with a focus on two-class problems via incremental learning [100]. The ensemble drift detection approach reduces the computational complexity of the updating process by selecting only a subset of newly available data and enables the timely tracking of ongoing patterns through feature vector selection [101]. CD has been addressed in financial time-series prediction to improve forecasting accuracy by employing ELM with explicit drift detection and online updating of the decision model when CD occurs in the data [4]. More importantly, ensemble-based methods provide a flexible and effective framework for CD detection in time-series analyses, enabling researchers to maintain model accuracy and improve forecasting performance.

In addition, various AI-based learners play a vital role in effective CD detection. For example, providing a diverse

framework for detecting CD in time-series data enables proactive adaptation and improves forecasting performance. Decision tree-based learners, such as XGBoost, k-NN, and FR, are effective in detecting sudden changes in data distribution and responding to incremental and abrupt drifts [59], [66]. Neural networks comprising RNN and MLP can learn complex temporal dependencies, making them suitable for detecting CDs, such as gradual and recurring drifts [48], [68], [118]. In addition, SVM-based learners focus on the decision boundary between classes, and SVMs can effectively detect shifts in the underlying data structure, which is particularly useful for sudden drifts. Critical issues related to CD have been addressed using incremental learning with SVM, which allows the acquisition of new knowledge without retaining previous knowledge [64]. PNN, another AI-based learner, has been used to analyze and investigate CD problems in models designed for fake review detection [72].

The proposed study selected 60 studies on CD detection and thoroughly investigated dataset sources to better understand the research results. The investigation revealed that some studies used real-world datasets, whereas others used synthetic datasets or a combination of both. Rahmani et al. [9] used real electronic health records to simulate data drift, covariate shift, concept shift, and a major event to study their effects on ML-based clinical sepsis prediction models. A novel uncertain label request strategy based on a variable least-confidence threshold vector was evaluated using a real network traffic dataset [62]. In addition, the proposed strategy attempts to address the challenges posed by the variable multiclass imbalance ratio and CD in network traffic classification. The Intel Berkeley Research Lab provided real temperature data from 54 Mica2Dot sensors that contained weather plates. The proposed method was evaluated using the fog-DeepStream method, which integrates three fields of study: signal processing, CD, and deep neural networks [63]. In a similar study, the authors used the real NTUST_EE_Lobby dataset with CD and reused the knowledge learned from the VOC dataset to test their proposed counterparts for streaming image analytics models [51]. Kaminskyi et al. [61] used synthetic datasets that contain Changing Sine Sudden (CS_{sudden}) and Changing Sine Incremental ($CS_{incremental}$), to effectively capture drifts in multivariate data streams.

Mehmood et al. [46] also used both synthetic and real datasets to detect CD and adapt it to smart city applications. Wang et al. [47] also used both to test a novel quadruple-based approach for understanding CD in a data stream framework using two real-world datasets as well as artificial datasets. The proposed method detects and predicts the start points of the CD. The effectiveness of incremental support vector ML in handling increasing and changing data (data drift) was evaluated using five industrial datasets, including clean and credit data [64]. Finally, a dataset from the Australian New South Wales (Electricity Market), including spam and mixed abrupt changes, was used to test an error-based weighted averaging ensemble model for adaptation to CD [65].



FIGURE 12. Lessons learned and best practices for concept drift mitigation in time-series applications.

Table 12 summarizes the evaluation metrics and datasets for CD detection in the selected studies.

In general, each learner and algorithm have strengths and weaknesses in terms of CD, handling, and adaptation. This could include the computational complexity of certain algorithms, sensitivity of specific methods to outliers, or the inability of some methods to detect certain types of drift (e.g., gradual drift). Table 13 summarizes the state-of-the-art studies, along with their contributions, strengths, and weaknesses.

VIII. SUMMARY OF LESSONS LEARNED AND BEST PRACTICES

All in all, this SLR examines recent research contributions related to CD detection in time series applications. The study of CD in time-series applications is important because of the increasing number of time-series and data streams and frequent changes in their underlying statistical properties over time. Detecting and adapting to these changes is required to maintain the performance of predictive models. This section summarizes the lessons learned and the best practices. Figure 12 presents a graphical representation of the summary of lessons learned and best practices for CD detection in time-series applications.

A. COMPUTATIONAL COST AND EFFICIENCY

A critical lesson from the literature is the necessity of predictive performance with computational cost and efficiency, particularly in real-time applications based on time-series data. The literature consistently prioritizes the development of more computationally tractable adaptation techniques due to the significant overhead associated with complete model re-training in response to CD. These include incremental learning methods, such as incremental adaptation using SVM for drift data, which adapts without re-training from scratch, and partial-update mechanisms that modify only specific model components [64].

Additionally, innovative architectural solutions such as the TriLS approach effectively address lightweight model tuning by transferring the computationally demanding tasks

of drift detection and model rebuilding to the cloud [18]. This process leads to a notable decrease in the computational load, memory requirements, and communication overheads of the gateway. At the most constrained level, TML-CD solutions are being developed to facilitate on-device adaptation while operating under stringent memory and energy constraints [50]. The collective approaches illustrate a notable lesson for developing lightweight, efficient, and architecturally informed solutions that facilitate practical CD handling for real-world implementations.

B. SELECTING APPROPRIATE WINDOW SIZES

In window-based methods, the selection of an appropriate window size is important and presents challenges for the CD. However, fixed-size windows can encounter difficulties with sudden drifts, whereas adaptive windows introduce additional complexities [70]. However, adaptive windowing techniques have been identified as the most effective approach. For example, ADWIN dynamically adjusts the window size based on variations in the observed data. Additionally, it operates by controlling a window and decreasing its size when the statistical differences between sub-windows surpass the specified confidence level, indicating a drift.

C. HANDLING DIFFERENT DRIFT TYPES

Effective detection of CD requires the implementation of techniques specifically designed for different types of drifts. Sudden drift, defined by abrupt shifts in concept, often requires immediate model re-training [88]. Incremental learners may effectively manage gradual changes, whereas abrupt drifts may necessitate rapid adaptation or model resetting. Recurring concepts are enhanced by mechanisms that enable the recall or reactivation of information from the previous model. Additionally, incremental drift, characterized by continuous and consistent change, may be enhanced through online learning or techniques that facilitate incremental updating of knowledge bases [89]. The OAR-DLSTM method assigns prediction tasks to multiple sub-models based on recurring concepts to manage hybrid recurring drifts in industrial processes [71]. The lesson learned is that it is essential to differentiate these types of drifts to select the most suitable and resource-efficient adaptation mechanism, thereby optimizing model performance in time-series environments.

D. DIFFERENTIATING DRIFT FROM NOISE

Differentiating between the actual CD and random fluctuations or outliers is a key challenge, as misidentification leads to excessive model updates and instability. For example, multiscale structure of MDDT was designed to effectively filter the noise present in drift indicators [82]. This study additionally confirmed that a significant challenge in CD detection is the accurate differentiation between actual concept shifts and mere noise or random fluctuations present in time-series data. Excessive sensitivity to noise may result in frequent, pointless model updates and possibly degrade

model performance [83]. The model may become outdated if it is overly resistant to noise, which may delay the identification of the actual drift.

E. DIVERSITY AND MANAGEMENT OF ENSEMBLES

Ensemble learning combines many base learners to provide a robust paradigm for detecting CD in time-series data. Recent research indicates that, in ensemble approaches, maintaining diversity among base learners is important for optimal performance, especially in the context of recurring drifts. Algorithms such as HDWM using “seed” learners preserve diversity [88]. A diverse ensemble consisting of models trained on distinct data subsets or employing various algorithms can collectively sustain high performance by changing CD [76]. Additionally, in ensemble methods, preserving diversity among base learners is essential for ensuring robustness and adaptability, particularly in the context of recurring drifts. Thus, diversity is not an appropriate replacement for accuracy. The optimal level of diversity depends on the speed and severity of drift.

F. THOROUGH EXPERIMENTAL EVALUATION

An experimental evaluation is necessary to validate CD detection methods. This process involves using various datasets, including both synthetic and real-world time series, to evaluate performance across different types of drifts [76]. The necessity for thorough experimental analysis employing various datasets, both synthetic and real-world, to simulate different types of drift and data conditions (imbalance and noise) is consistently emphasized for the validation of new methods [48]. Multiple metrics can be considered during evaluation, including accuracy, F1-score, detection delay, false alarm rate, computational complexity, and memory use. The best practices revealed that a comprehensive evaluation across CD models is important for assessing a technique in time-series forecasting.

G. SYSTEM ARCHITECTURE FOR DRIFT HANDLING

An effective system architecture for handling CD in time-series data typically consists of several interconnected modules, which are considered the best practices. The interconnected modules incorporate novel edge architectures, including a hybrid cloud-edge continuum, which are proposed to support AI workloads and facilitate automated CD handling for smart city applications [46]. Based on resource availability, these architectures can enable workload migration on demand (e.g., model retraining). Additionally, hybrid cloud-edge architectures can move computationally intensive drift detection and model rebuilding to the cloud, whereas edge devices handle local lightweight adaptation [18].

H. REAL-WORLD VERSUS SYNTHETIC DATA

Evaluating CD detection methods requires careful analysis of data sources, especially distinguishing between real-world and synthetic data. Real-world data are extremely important

TABLE 12. Summarizes of the evaluation metrics and datasets for CD detection from the selected studies.

Evaluation metrics	Datasets and Repository	Datasets	
		Real	Synthetic
MAE, RMSE, MAPE [46]	Approximately 419 sensors stream data from a smart campus in Finland (University of Oulu).	✓	✓
Accuracy, F1 score [47]	Real-world data: Power supply, Posture Synthetic: Circle, Sine, Hyperplane, RandomRBF.	✓	✓
AUROC [9]	Electronic health records (EHR) of inpatients at four hospitals in the United States.	✓	
AVF, CBA [62]	Network traffic datasets: Entry10, KDDTrain5, UDP_NoPort, TCP_NoPorts, NIMS1, URL_All, URL_BestFirst, URL_Infogain. (https://github.com/sunnyweike/MicFoil)	✓	
RMSE, MSE, MAE [63]	Temperature data were obtained from the Intel Berkeley Research Laboratory.	✓	
Precision, Recall, F1 score [49]	Network intrusion dataset released in 2018 and APT Command and Control dataset. (https://github.com/SeldonIO/alibi-detect?tab=readme-ov-file#datasets)	✓	
Accuracy F1 score, Precision [61]	The datasets were designed to generate two types of drift: sudden and incremental.		✓
Accuracy [64]	Five industrial datasets, including clean data, credit data, and waveform data with high dimensions.	✓	✓
Accuracy, Precision, Recall and F1 score [65]	Two real-world datasets and one synthetic dataset, which are electricity, spam, and mixed-abrupt.	✓	✓
Accuracy, Precision, Recall, F1 score [67]	IoT-based intrusion detection on distributed smart space orchestration system (DS2OS) traffic traces datasets.	✓	
Accuracy, Precision, Recall, F1 score, G-Mean [69]	The data source contains 140,758 vulnerabilities disclosed between 1988 and 2020.	✓	
MASE [8]	UCI Electricity Load Dataset. (https://archive.ics.uci.edu/dataset/321/electricityloadaddiagrams20112014)	✓	✓
Accuracy [80]	Bot-IoT dataset, including the popular synthetic datasets SEA Concept and UG-2C-5D. (https://github.com/leslin1010/SeaConceptDataset)	✓	✓
Accuracy, Precision [51]	NTUST_EE_Lobby dataset	✓	
AUC, F1 score, MCC [58]	Google dataset and disk failures in Backblaze dataset.	✓	

for validating the robustness, practicality, and applicability of CD detection methods [72]. Synthetic data enable precise control over drift characteristics [80]. This allows for a systematic evaluation of detection accuracy, latency, and false positive rates under specified conditions. However, real-world drifts often differ from simulated drifts. Additionally, it is important to evaluate the methods on various datasets, including real-world and synthetic datasets with known drift types. The lesson learned is that both data types are important because synthetic data allow rigorous and controlled experimentation, whereas real-world data demonstrate practical applicability and unforeseen challenges.

IX. CONCLUSION AND FUTURE DIRECTIONS

In conclusion, forecasting time-series applications rely on the output from the ML model after training the data, which becomes more challenging because of the CD problem. This study examines the current state-of-the-art methods that detect and adapt CD in time-series data for regression and classification learning tasks. These studies have enabled the development of a taxonomy for this area. After critically reviewing the existing literature, this study developed a taxonomy and presented an experimental evaluation of the proposed roadmap for CD detection. Additionally, this study summarized, categorized, mapped, and surveyed the existing literature on drift detection in time-series data by developing RQs based on existing empirical studies. The PRISMA 2020 method was used to conduct the survey, and 60 studies

published between 2013 and 2024 were carefully selected and assessed. This SLR also confirms that using AI-based learners to detect, handle, and adapt to CD in time-series scenarios is a promising approach.

However, the SLR has some limitations. First, the study analyzed 60 research articles selected based on a search strategy that involved searching multiple databases, such as SCOPUS, Science Direct, IEEE Xplore, Web of Science, MDPI, and ACM. Therefore, because the search criteria focused on journal papers published between 2013 and 2024, some relevant studies were excluded from the review. Studies conducted in languages other than English were excluded. Second, most of the currently proposed drift detection, handling, and adaptation methods have not been solved. However, reliance on simulation datasets may not fully capture real-world scenarios. Moreover, several studies have tested the proposed method using a single dataset, which may limit generalizability. In addition, comparisons with active approaches from state-of-the-art studies are limited. Finally, this study was unable to investigate other approaches to CD detection, such as similarity and dissimilarity-based methods, using time-series data.

This study recommends that further research investigate existing algorithms for AI-based learners by introducing next-generation approaches for detecting CD in time-series and data streams. The analysis of databases other than SCOPUS, Science Direct, IEEE Xplore, Web of Science, ACM, and MDPI is also recommended. The following

TABLE 13. Summary of state-of-the-art studies and their contributions, strengths, and weaknesses.

Algorithms/Leaners	Contribution	Strengths	Weaknesses
ADWIN, KSWIN, LSTM [46]	Presented an innovative computational framework for CD detection in smart settings.	Identifies CD as a key issue in real-world time-series forecasting.	Additional testing with larger datasets and false positive scenarios is required to evaluate the robustness of the proposed method.
QuadCDD, LSTM [47]	Presents a novel framework called QuadCDD, which provides a comprehensive understanding of CD in data streams.	Enhance the stability and performance of the model when faced with CD.	There is insufficient discussion of specific decision-making or adaptation techniques based on drift type.
XGB, RNN [9]	Examines the effect of data drift, specifically the occurrence of concept and covariate shifts, on sepsis prediction models by analyzing electronic health record (EHR) data.	Employs simulations to investigate different instances of data drift and analyze their influence on model performance.	Relying on simulations that do not fully capture real-world complexities.
CUSUM, EWMA, PHT, LSTM, DNN, ANN [63]	A framework for effective CD and data stream analytics in fog computing called Fog-DeepStream is presented.	The proposed method combines signal processing, CD detection, and DL to handle resource-constrained scenarios.	The relationship between the transmission cost and drift detection accuracy was not investigated.
ADWIN [18]	Present a novel TriLS system for drift-aware time-series modeling in the IIoT environment.	Detects and handles CD using an efficient real-time prediction approach.	The evaluation used only two datasets, which may limit the generalizability of the results.
ADD, FCN, RNN [68]	Proposes Meta-ADD, a novel framework for detecting CD in time-series data streams.	Meta-ADD automates drift detection and classification to reduce human bias and effort in hypothesis testing.	Meta-learning and active learning techniques incur computational costs.
MDDT, CUSUM, k-NN [82]	Present a novel approach to detect abrupt CDs in time-series data with fluctuating feature values.	The multiscale structure significantly reduces computation time compared to traditional passive techniques.	Limited comparison with active approaches.
DDM, HDDM [87]	Present a novel framework (ESCR) for handling CD in data streams using a semi-supervised approach, which can be effective when labeled data is scarce.	Using semi-supervised learning addresses the challenge of limited labeled data, which is common in CD scenarios.	Limited details about base classifiers and outlier handling.
GA, NB [52]	Proposes a CD detection method for spam filtering based on the Statistical Test of Equal Proportions (STEPD).	CD detection is employed to potentially improve spam filter adaptability.	Lacks details about how STEPD specifically detects CD in email content distribution.
AL-ELM, LSTM [107]	Introduces a dual-learner ensemble strategy (alternating learners' framework) to detect CD in a time-series.	Effectively detects and predicts concept changes, both abrupt and gradual.	Lack of exploration of effectiveness on real-world datasets beyond simulated power plant data.
HDDM, NB [96]	Presents innovative techniques for identifying CDs in online learning using moving and weighted moving averages.	The methods are efficient and require few computational resources.	This study focused on two specific weighting schemes: moving averages and weighted moving averages. Investigating other weighting options may be beneficial.
Learn++.CDS [99]	Introduces an ensemble-based approach (Learn++.CDS) for handling CD in imbalanced time-series data.	Addresses the growing real-world challenge of CD caused by imbalanced data.	Higher computational complexity than existing methods, particularly Learn++.NIE, because of sub-ensemble creation.

research directions are recommended based on the SLR findings.

- 1) Most existing methods in the literature require training models after drift detection, which can be

time-consuming and disruptive in real-world scenarios [14], [19]. Adaptive learning models that can respond to CD without explicit training and ensure optimal performance are being developed. For example,

TABLE 14. PRISMA 2020 checklist.

Heading	Item	Section
Title	1	Title
Abstract	2	Abstract
Introduction	3	Section I
	4	Section I
Methods	5	Section III. B
	6	Section III. B
	7	Section III. A
	8	Section III. B
	9 – 15	Not applicable
Result	16a	Figure 4
	16b	Not applicable
	17	Section IV
	18 – 22	Not applicable
Discussion	23a	Not applicable
	23b	Section VII
	23c	Section VII
	23d	Not applicable
Other information	24a – c	Not applicable
	25	Acknowledgment
	26	Declaration of interests
	27	Not applicable

online learning algorithms continuously update models based on new data.

- Several existing approaches require labeled data to detect drift, which can be difficult or expensive [119]. Exploring unsupervised and semi-supervised learning approaches for CD detection that use a small amount of labeled data to guide the detection process in scenarios with limited labeled data.
- The distinction between correlation and causation remains a significant challenge in CD identification [120]. Future studies should investigate causal representation learning methods, such as causal discovery and counterfactual reasoning, to enhance the robustness and generalizability of CD models.
- A single approach may not be effective in all scenarios. Hybrid approaches that combine several detection methods are potentially transformative. For example, combining statistical methods with windowing techniques can result in more robust detection.
- Future investigations should explore DL techniques that integrate complementary data sources, including sensor readings, images, and text, for CD detection. Advanced models, including multimodal transformers, can integrate these techniques to enhance the detection accuracy.

APPENDIX A

Tables 12 and 13.

APPENDIX B

The PRISMA 2020 checklist was obtained from https://www.prisma-statement.org/s/PRISMA_2020_checklist-k3f4.docx (accessed May 19, 2025). Table 14 presents the checklist used in this SLR, including the sections, numbers, and placements in the publication.

ABBREVIATIONS

This manuscript uses the following abbreviations:

AA	Average Accuracy.
ACC	Accuracy.
ACDDM	Accurate Concept Drift Detection Method.
ADWIN	Adaptive Windowing.
ADDM	Adaptive sliding window-based Detection Method.
ADD	Active Drift Detection.
AVF	Architectural Vulnerability Factor.
ANN	Artificial Neural Network.
AUC	Area Under the Curve.
AGE	Accuracy and Growth rate updated Ensemble.
AIOps	Artificial Intelligence for IT Operations.
CSDD	Cosine Similarity Drift Detector.
CDDA	Concept Drift Domain Adaptation.
CDBD	Confidence Distribution Batch Detection.
CUSUM	Concept Drift using CUMulative SUM.
CBA	Cost Benefit Analysis.
CNN	Convolution Neural Networks.
CDT	Change Detection Test.
CD	Concept Drift.
DT	Decision Tree.
DNN	Deep Neural Network.
DDM	Drift Detection Method.
ELM	ensemble Extreme Learning Machines.
XGBoost	eXtreme Gradient Boosting.
EDDM	Early Drift Detection Method.
FCN	Fully Convolutional Network.
EDDM	Early Drift Detection Method.
EWMA	Exponentially Weighted Moving Average.

ECDD	Exponentially Weighted Moving Average Concept Drift Detection.
ELM	Extreme Learning Machine.
FPR	False Positive Rate.
FP	False Positive.
FN	False Negative.
FHDDM	Fast Hoeffding Drift Detection Method.
FTRL-ADP	Follow-the-Regularized-Leader with Adaptive Decaying Proximal.
FSDD	Fisher Square Drift Detector.
FPDD	Fisher Proportions Drift Detector.
GNB	Gaussian Naïve Bayes.
G-Mean	Geometric Mean.
GA	Genetic Algorithm.
GMA	Geometric Moving Average.
GMM	Gaussian Mixture Model.
GRU	Gated Recurrent Unit.
HDDDM	Hellinger Distance Drift Detection.
HDDM	Hoeffding Drift Detection Method.
HDWM	Heterogeneous Dynamic Weighted Majority.
HW_DDM	Hybrid Weighting-based Concept Drift Detection Method.
KSWIN	Kolmogorov–Smirnov windowing.
k-NNs	k-nearest Neighbors.
KSWIN	Kolmogorov–Smirnov Windowing.
KELM	Kernel Extreme Learning Machine.
LR	Logistic Regression.
KRR	Kernel Ridge Regression.
LSTM	Long Short-Term Memory.
Learn++CDS	Learn++ for Concept Drift with SMOTE.
MAE	Mean Absolute Error.
MAPE	Mean Absolute Percentage Error.
MSE	Mean Squared Error.
MED	Median Absolute Error.
MOS-ELM	Meta-Cognitive Online Sequential Extreme Learning Machine.
MLP	Multi-Layer Perceptron.
MDDT	Multiscale Drift Detection Test.
MSPE	Mean Squared Prediction Error.
MASE	Mean absolute scaled error.
MCC	Matthew’s correlation coefficient.
NN	Neural Networks.
NB	Naive Bayes.
NDE	Number and Distance of Errors.
NRMSD	Normalized Root-Mean-Square Deviation.
OAUE	Online Accuracy Updated Ensemble.
PNN	Perceptron Neural Network.
PHT	Page-Hinkley Test.
RMSE	Root Mean Square Error.
RCA	Root Cause Analysis.
ROC	Receiver Operator Characteristic.
RNN	Recurrent Neural Network.
RF	Random Forest.

RHP	Rotating Hyper Plane.
RNN	Recurrent Neural Network.
RDCAL	Realtime Dynamic Concept Adaptive Learning.
RGF	Regularized Greedy Forests.
ROC	Receiver Operating Characteristic Curve.
SAM	Self-Adjusting Memory.
SVM	Support Vector Machine.
SD	Standard Deviations.
TP	True Positive.
TML	Tiny Machine Learning.
TML-CD	Tiny Machine Learning for Concept Drift.
WSTD	Wilcoxon Rank Sum Test Drift Detector.
YOLO	You Only Look Once.

DECLARATION OF INTERESTS

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this study.

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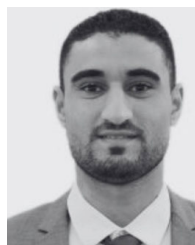


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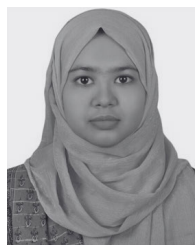
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